Shatter Clustering: Trajectory Step Acceptance (alpha_accept) Analysis

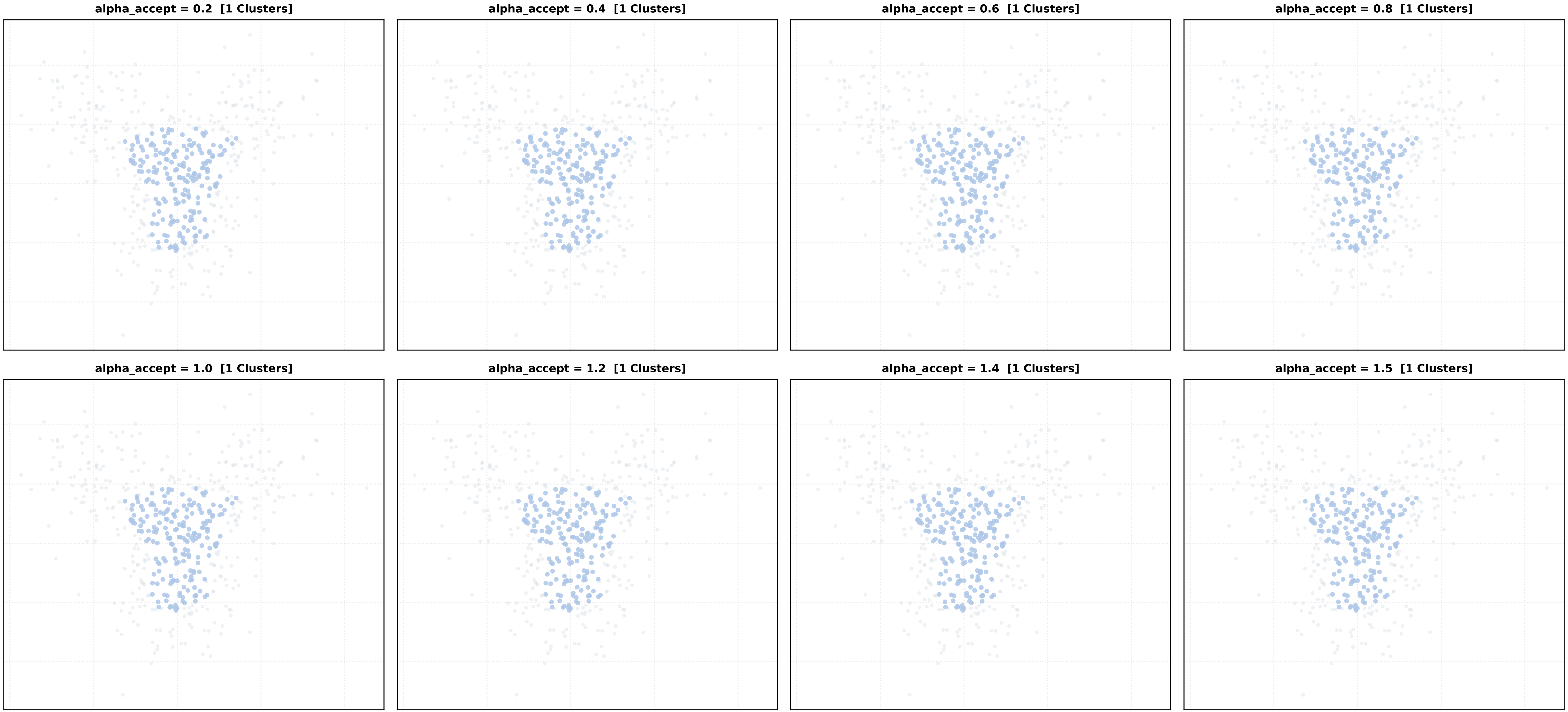
Sweep over $\alpha_{\text{accept}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.5\}$ *with Default Defaults*

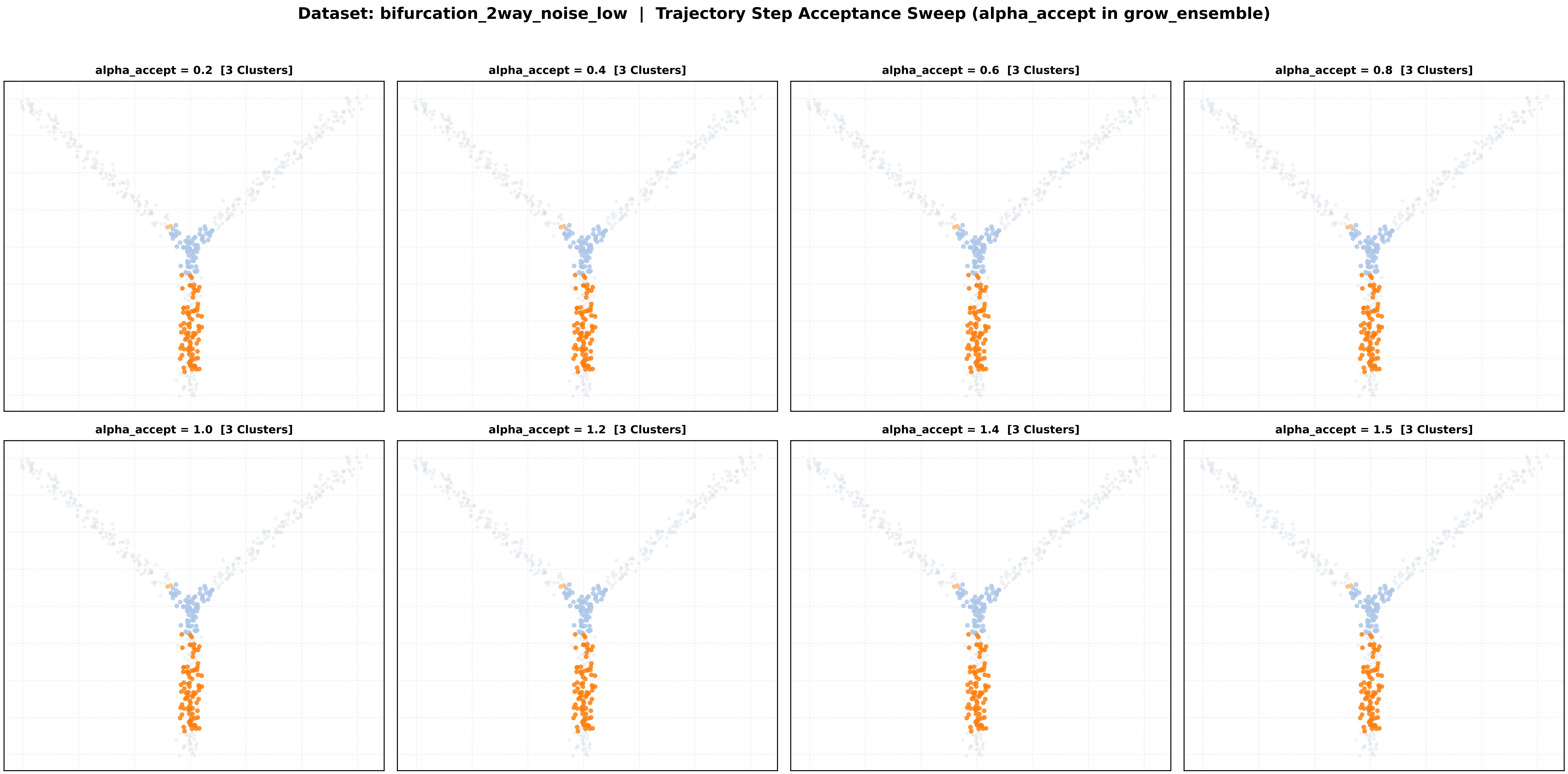
Experimental Configuration

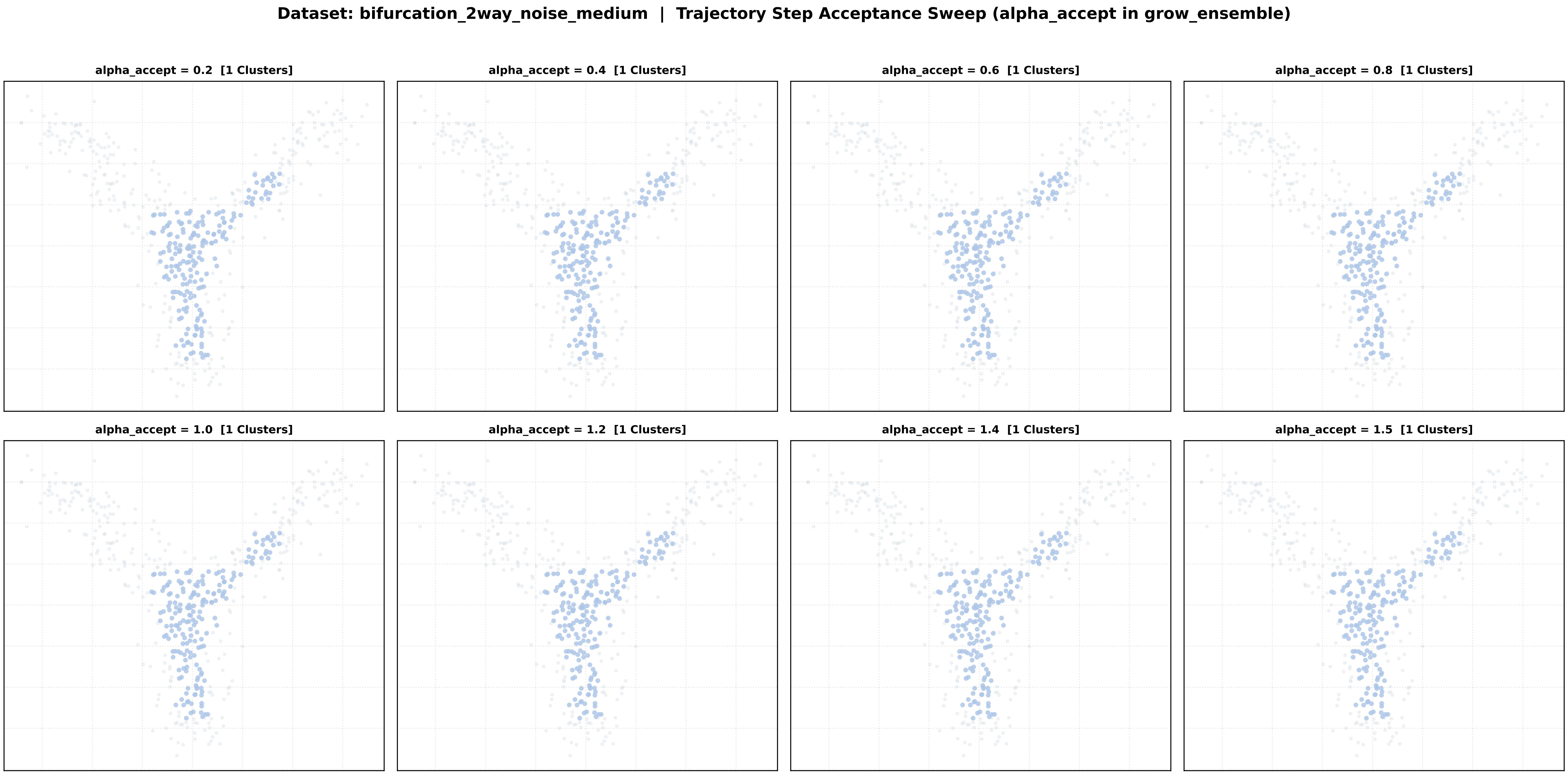
- Total Evaluated Datasets: 30
- Tested alpha_accept Values: [0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.5]
- Total Executions: 240 (8 runs per dataset)
- Fixed Default Parameters: alpha_MAD = 0.5, t_percentile = 0.15, quant = 0.15
- Target Output PDF: benchmark_report_alpha_accept.pdf

Parameter	Value / Range	Algorithmic Role
alpha_accept	{0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4	Log2 fold-change cutoff on H_rho step acceptance in grow_e
alpha_MAD	0.50 (Default)	MAD multiplier for surface growth compatibility
t_percentile	0.15 (Default)	Top 15% densest seeds
mean_dist_quant	0.15 (Default)	Quantile for adaptive search radius r
min_intersection	1 (Default)	Overlapping vectors to trigger merge

Dataset: bifurcation_2way_noise_high | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)




Dataset: bifurcation_2way_noise_low | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [3 Clusters]
alpha_accept = 0.4 [3 Clusters]
alpha_accept = 0.6 [3 Clusters]
alpha_accept = 0.8 [3 Clusters]
alpha_accept = 1.0 [3 Clusters]
alpha_accept = 1.2 [3 Clusters]
alpha_accept = 1.4 [3 Clusters]
alpha_accept = 1.5 [3 Clusters]


Dataset: bifurcation_2way_noise_medium | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [1 Clusters]
alpha_accept = 0.4 [1 Clusters]
alpha_accept = 0.6 [1 Clusters]
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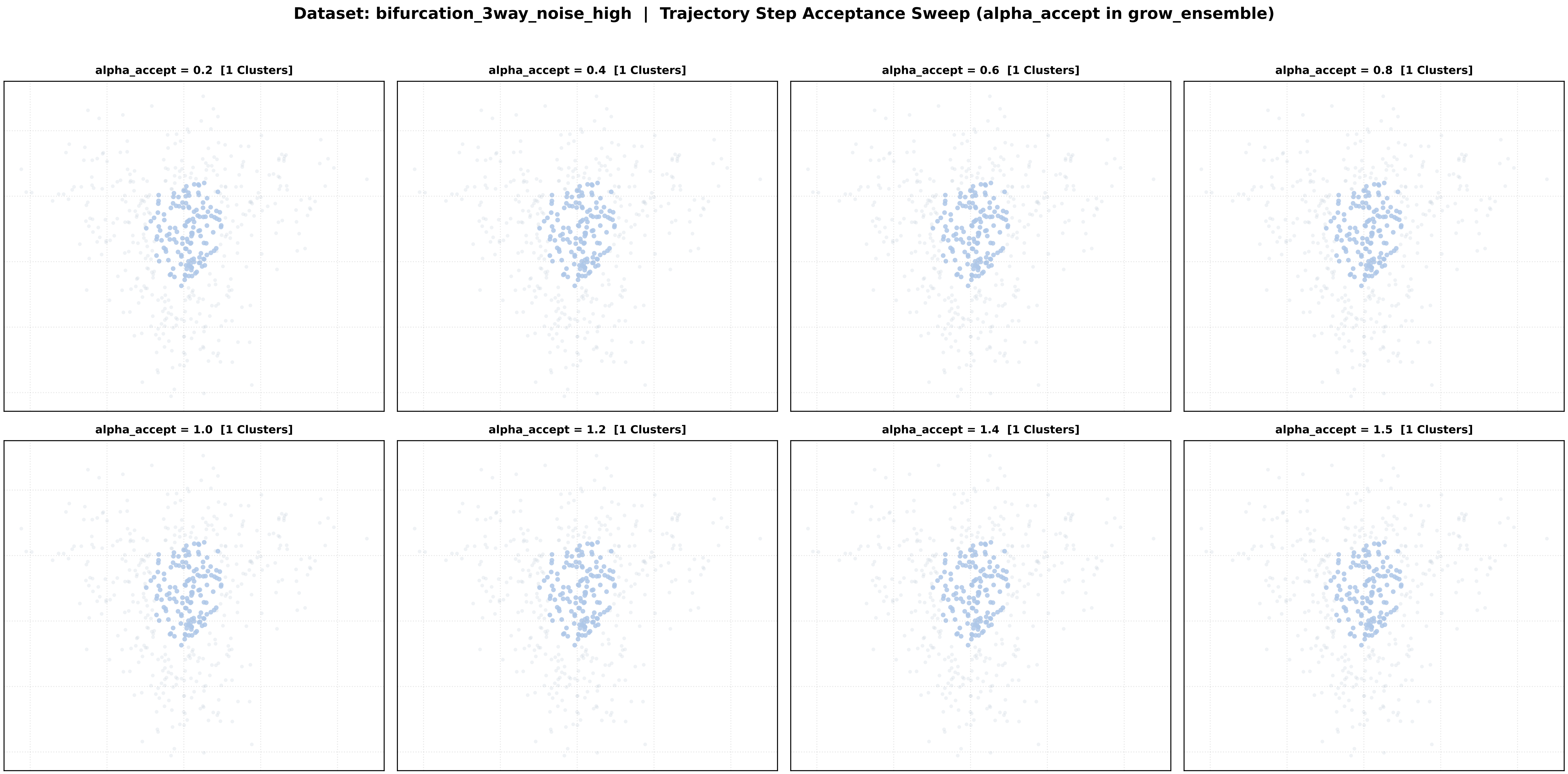
Dataset: bifurcation_3way_noise_high | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [1 Clusters]
alpha_accept = 0.4 [1 Clusters]
alpha_accept = 0.6 [1 Clusters]
alpha_accept = 0.8 [1 Clusters]
alpha_accept = 1.0 [1 Clusters]
alpha_accept = 1.2 [1 Clusters]
alpha_accept = 1.4 [1 Clusters]
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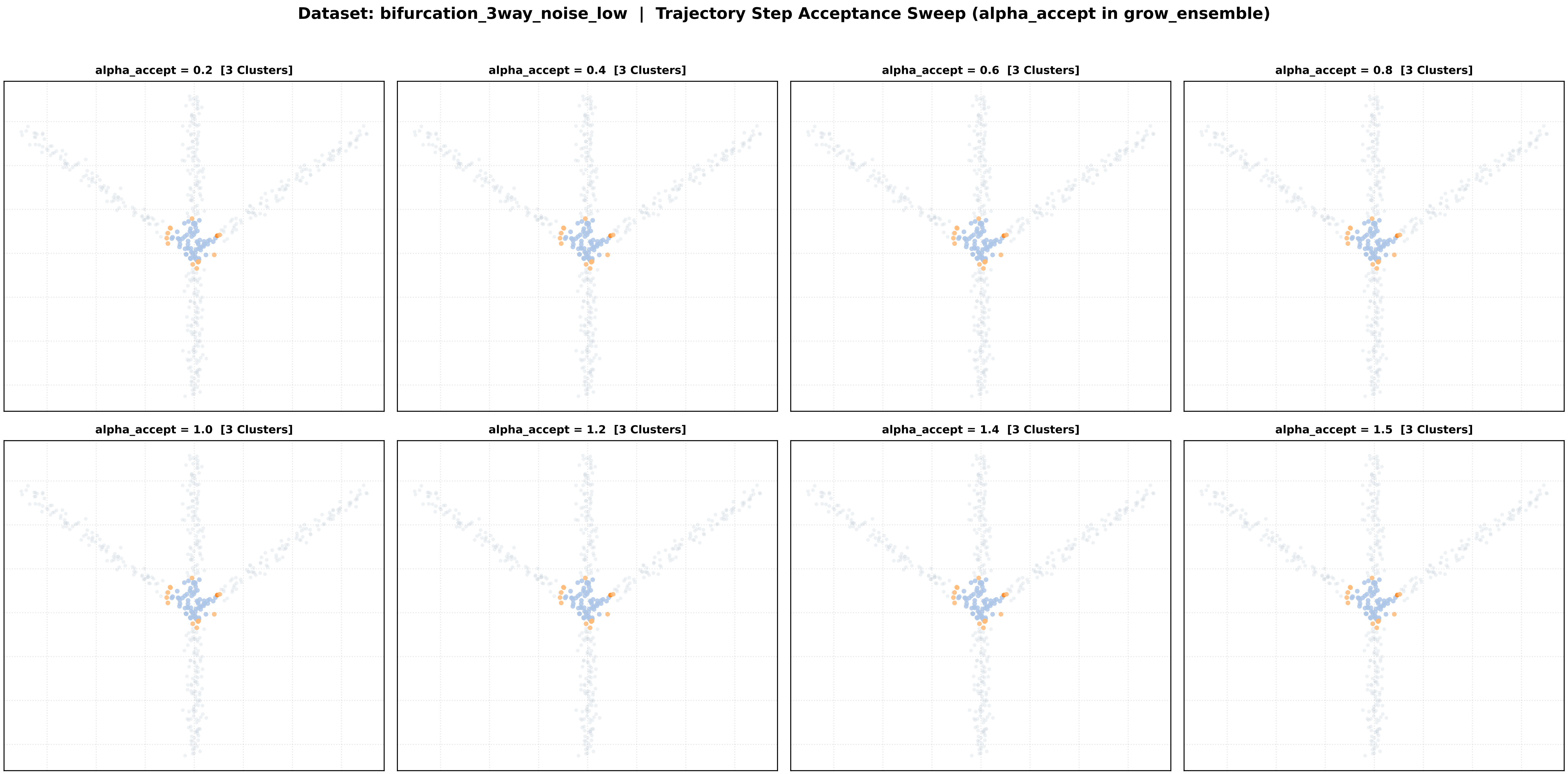
Dataset: bifurcation_3way_noise_low | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [3 Clusters]
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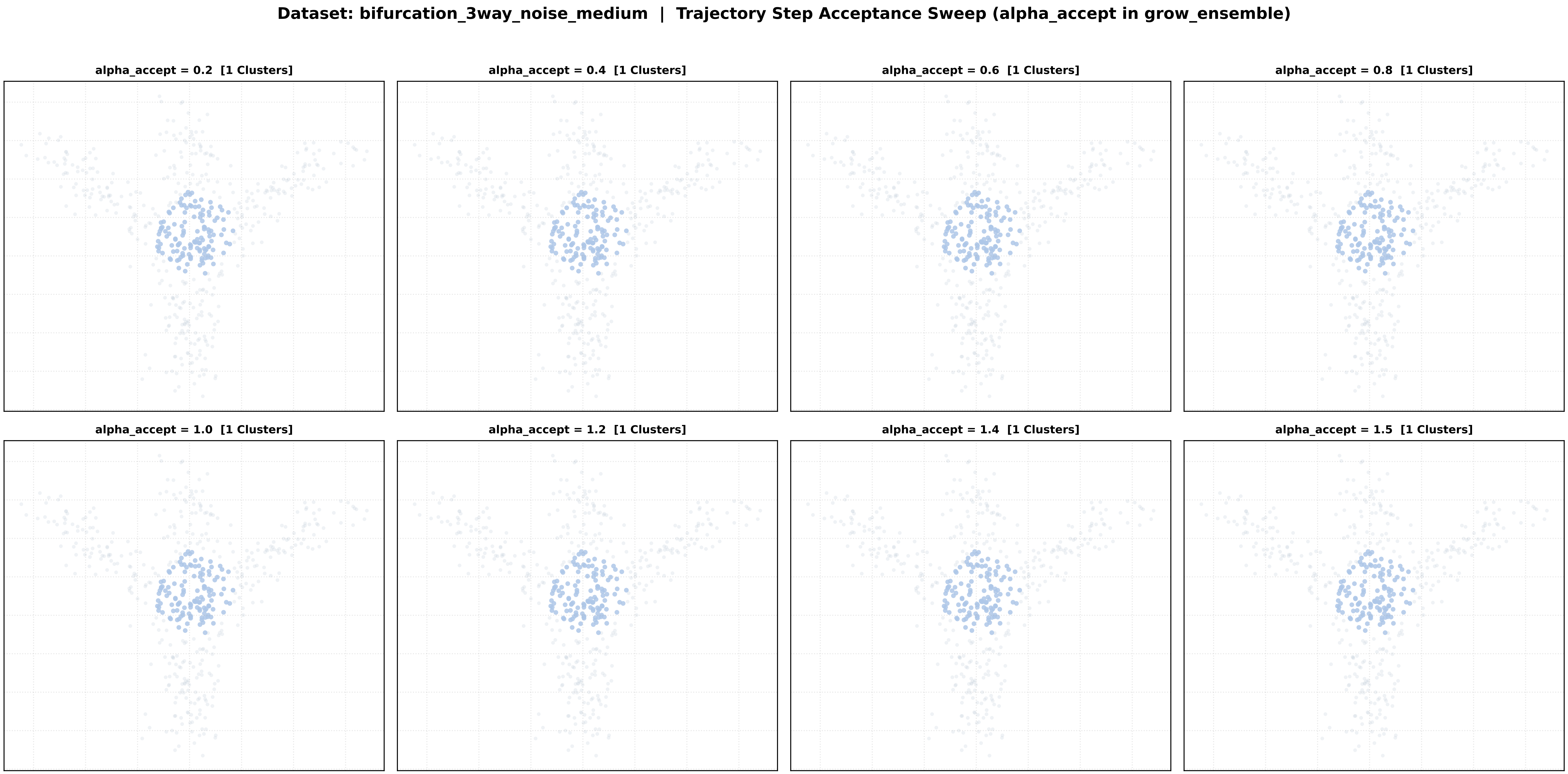
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alpha_accept = 0.2 [1 Clusters]
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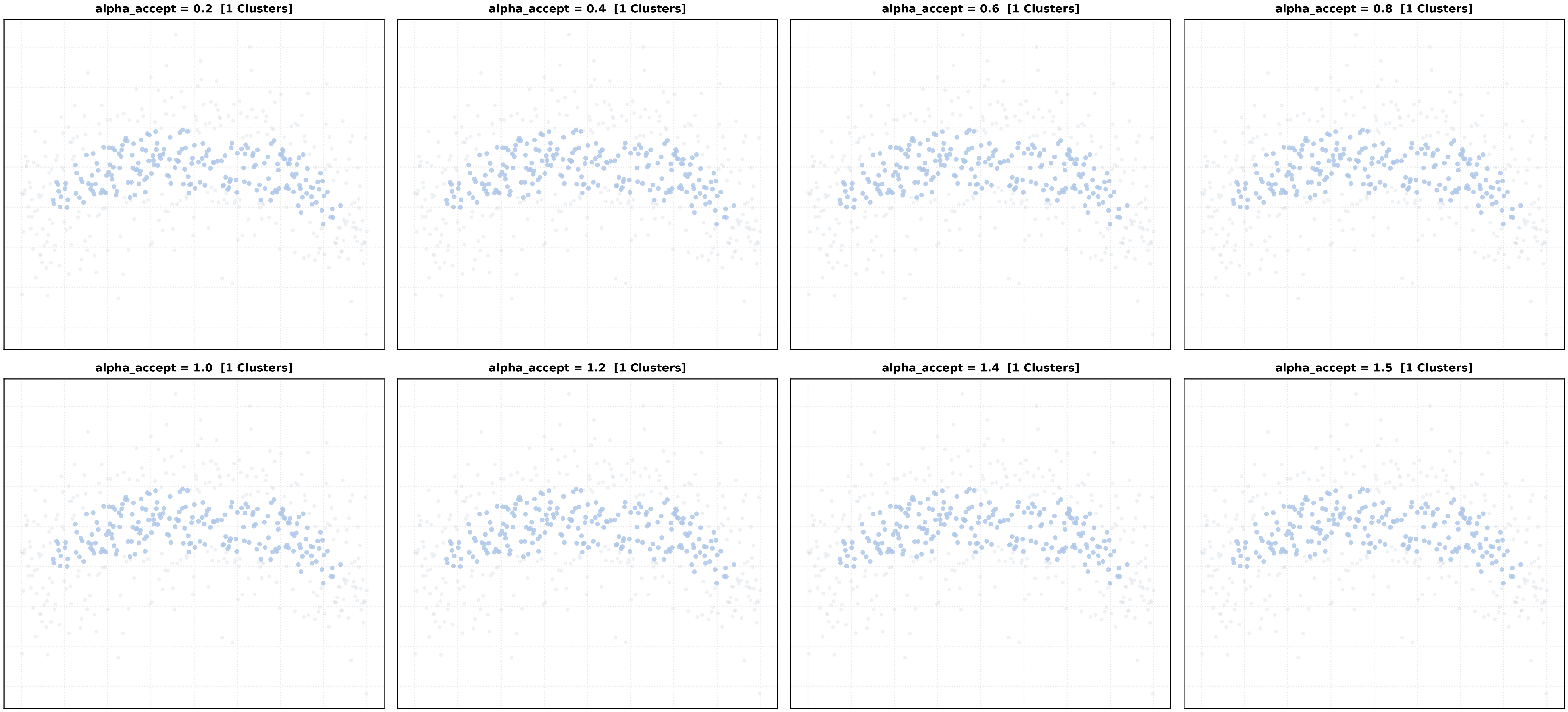
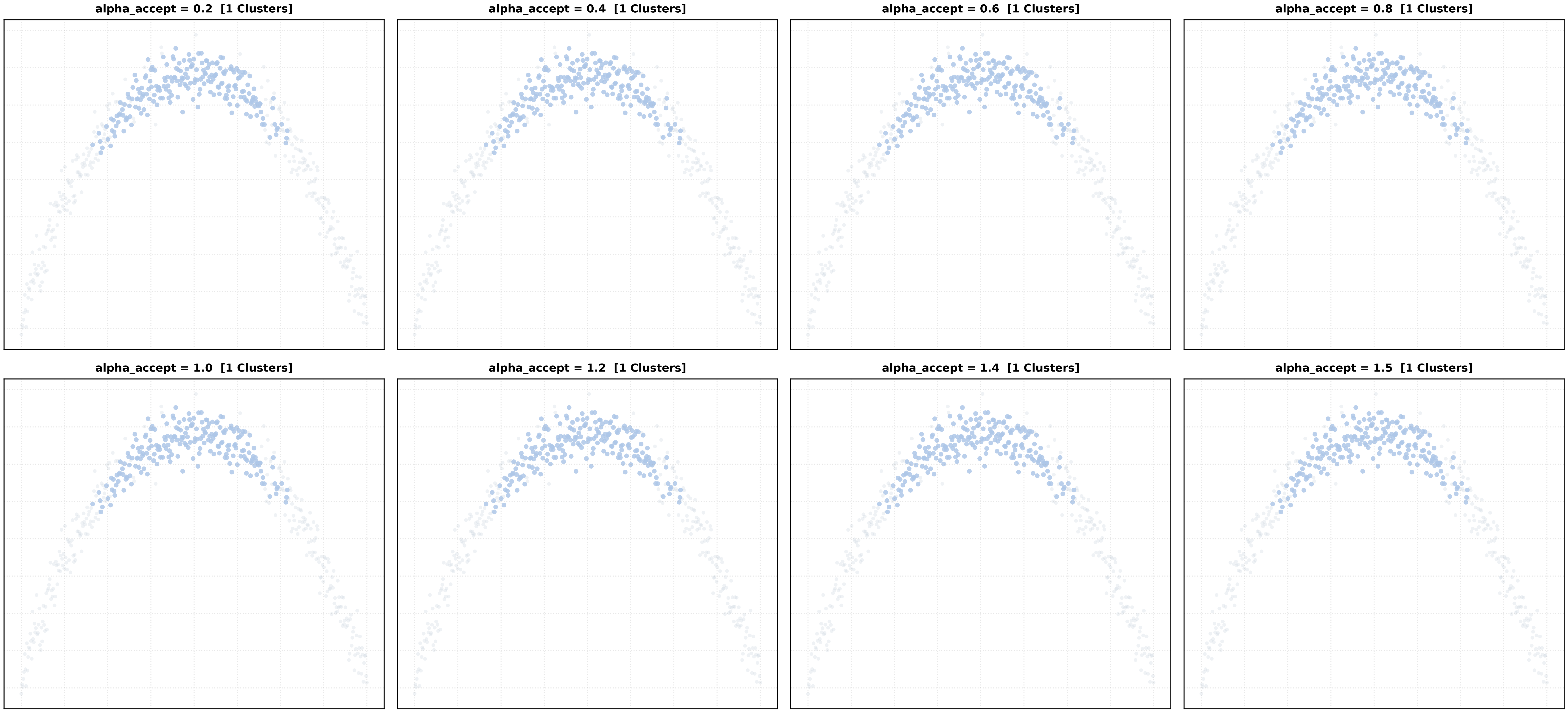
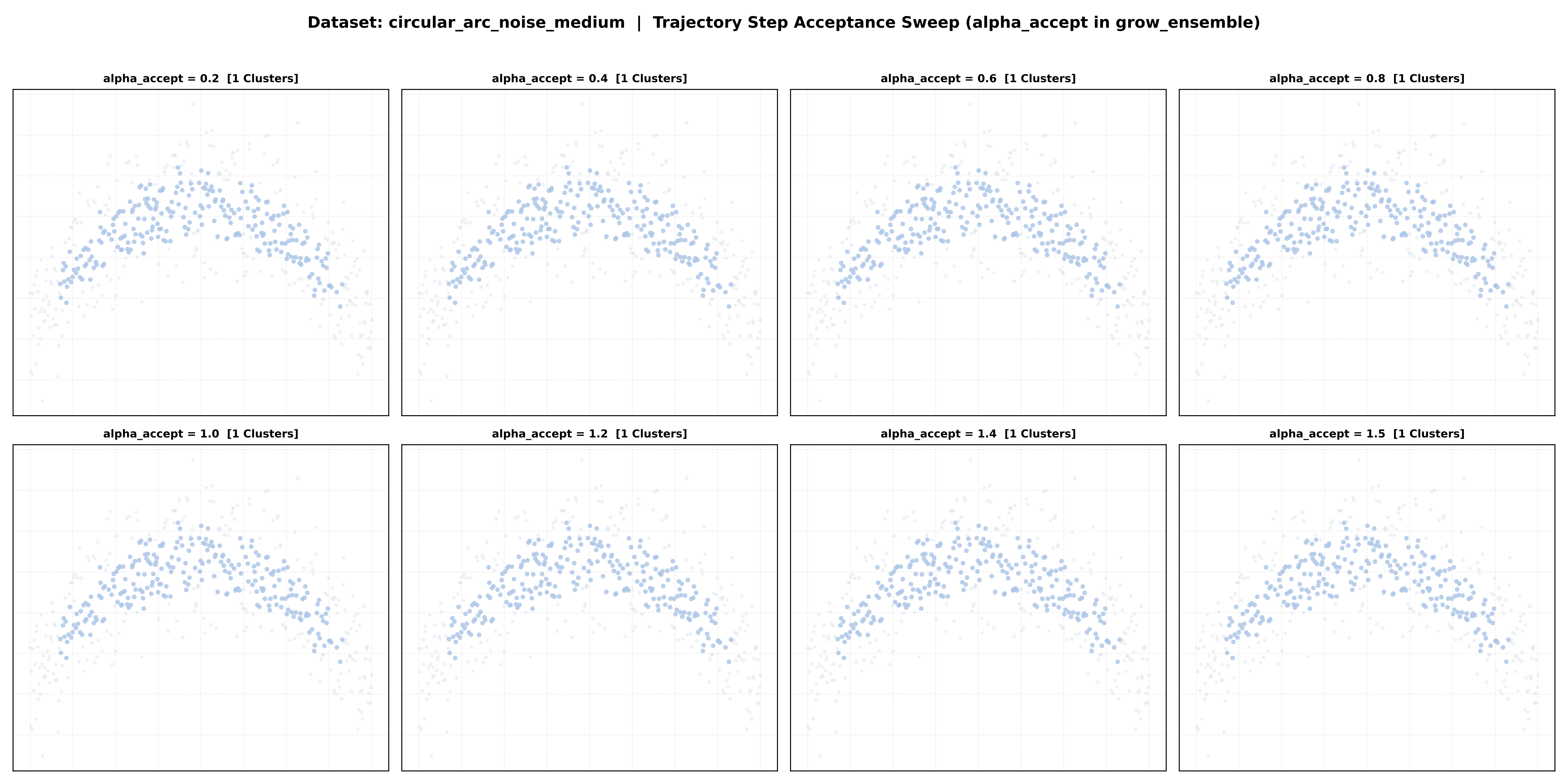
Dataset: circular_arc_noise_high | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)



Dataset: circular_arc_noise_low | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)




Dataset: circular_arc_noise_medium | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [1 Clusters]
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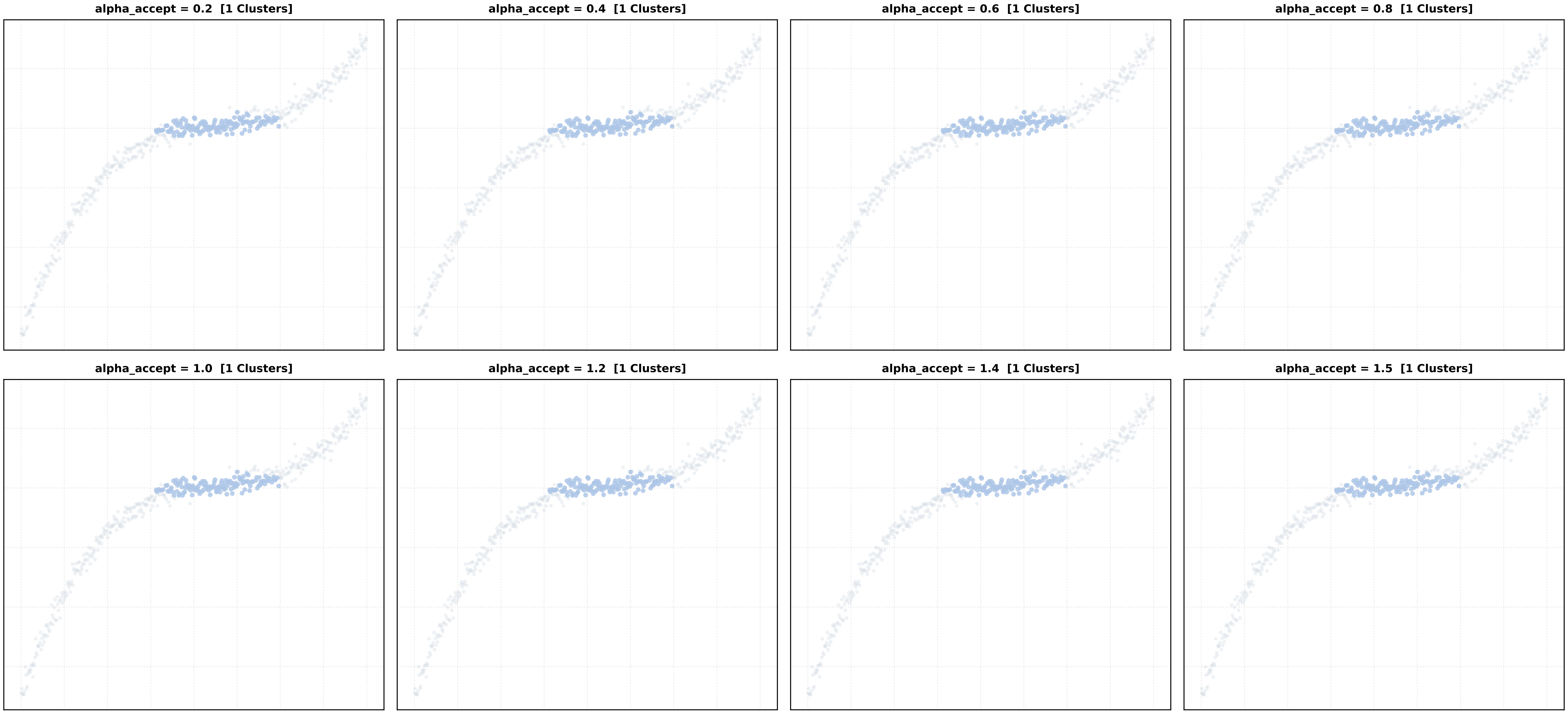

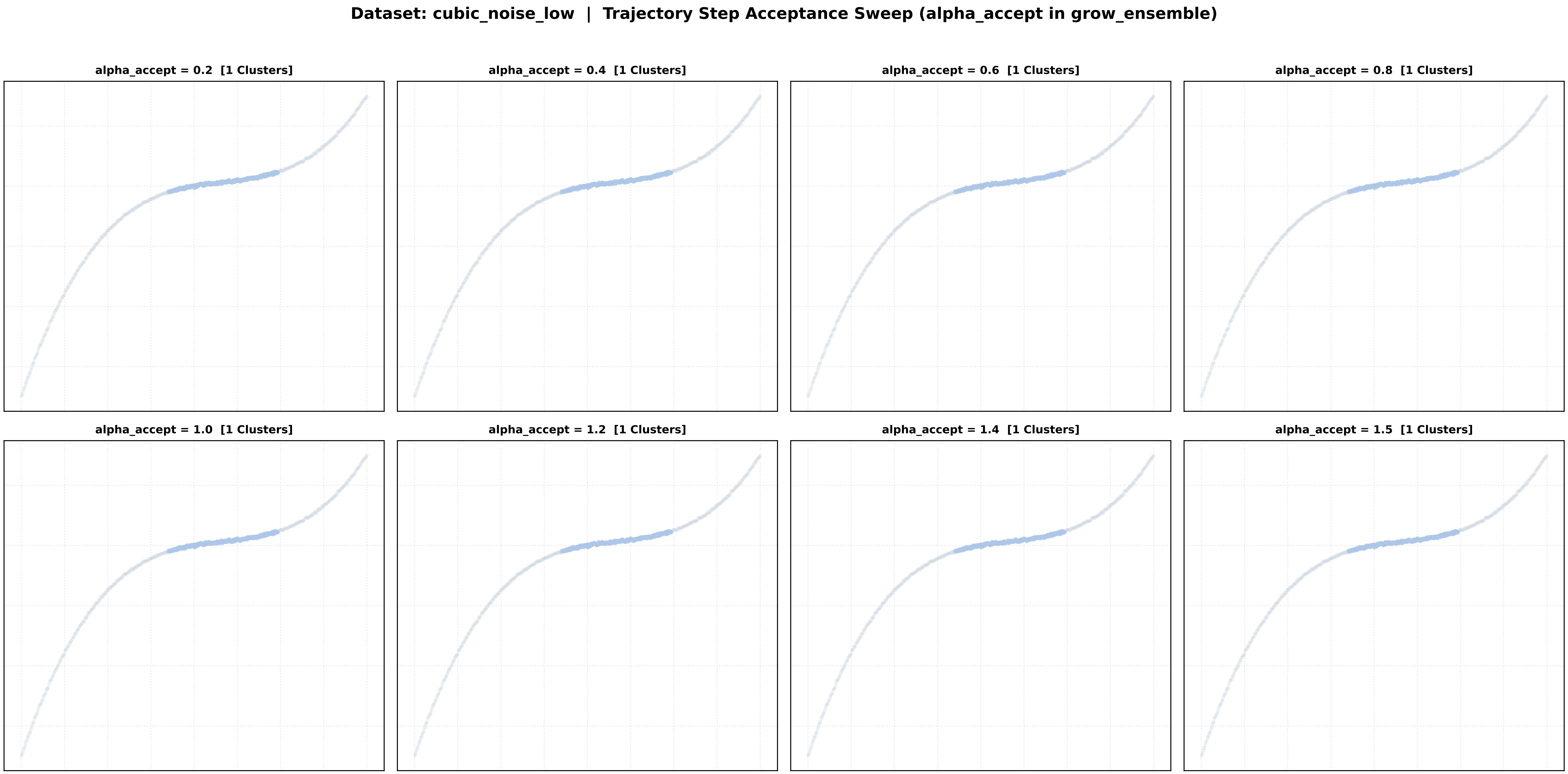

Dataset: cubic_noise_low | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [1 Clusters]
alpha_accept = 0.4 [1 Clusters]
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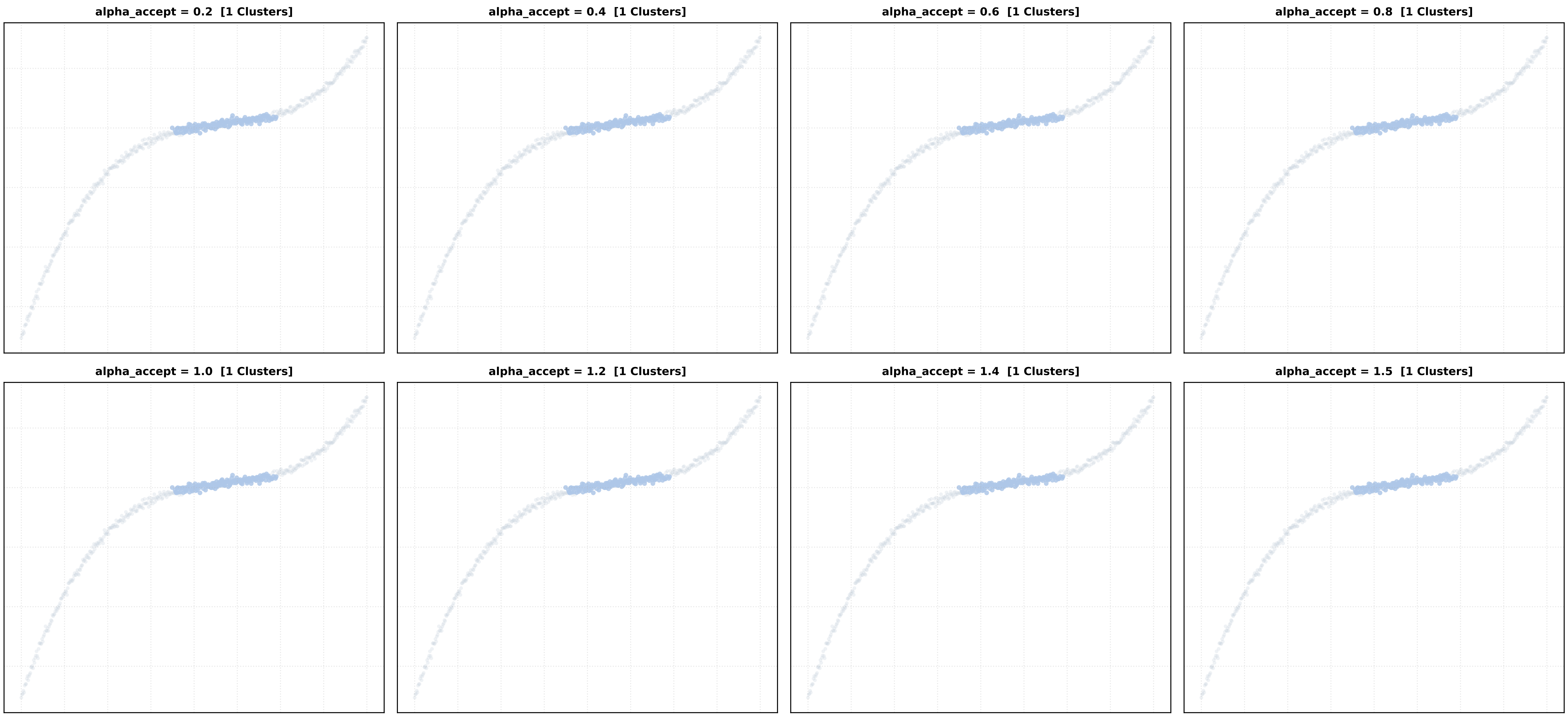
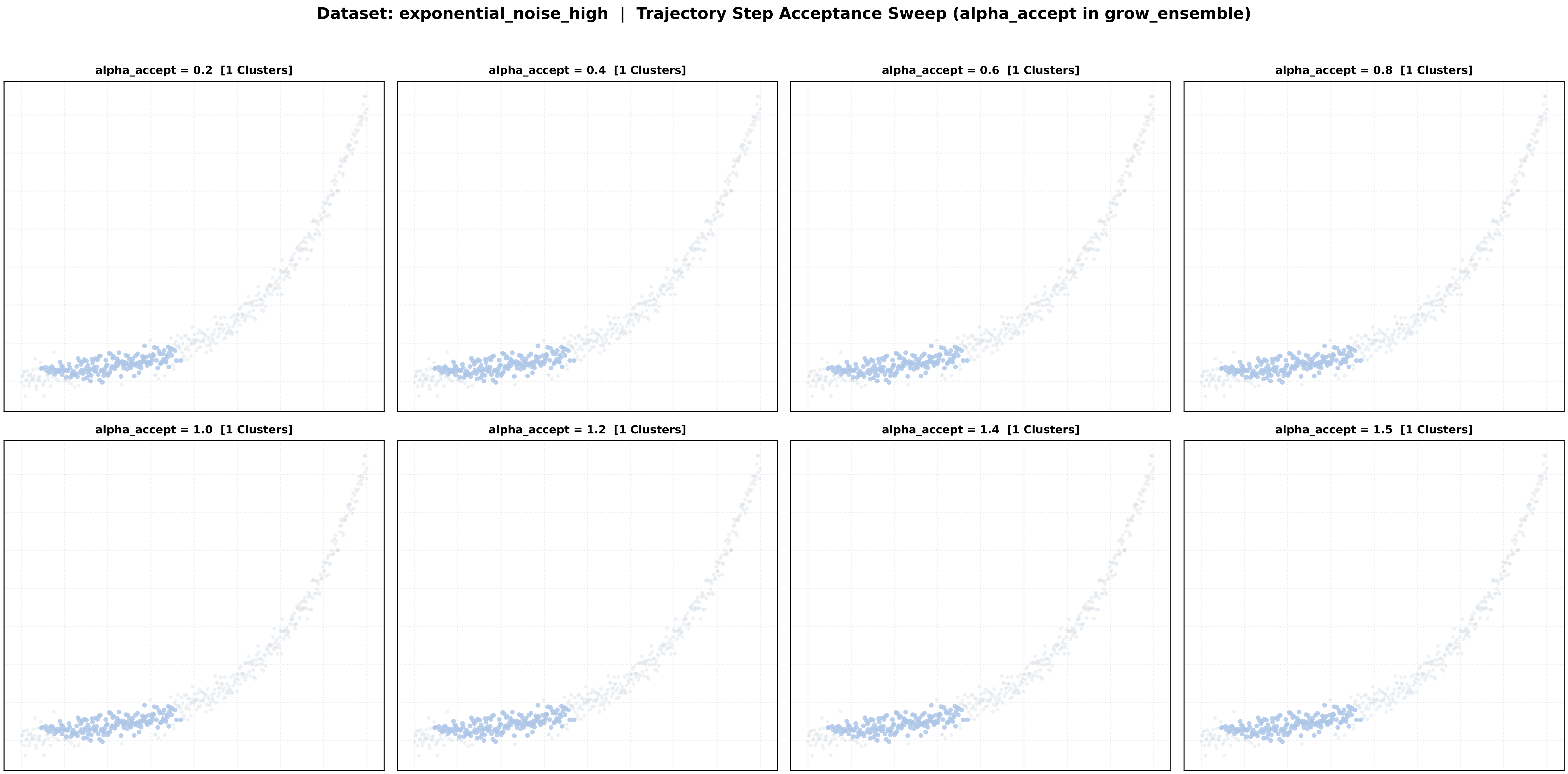
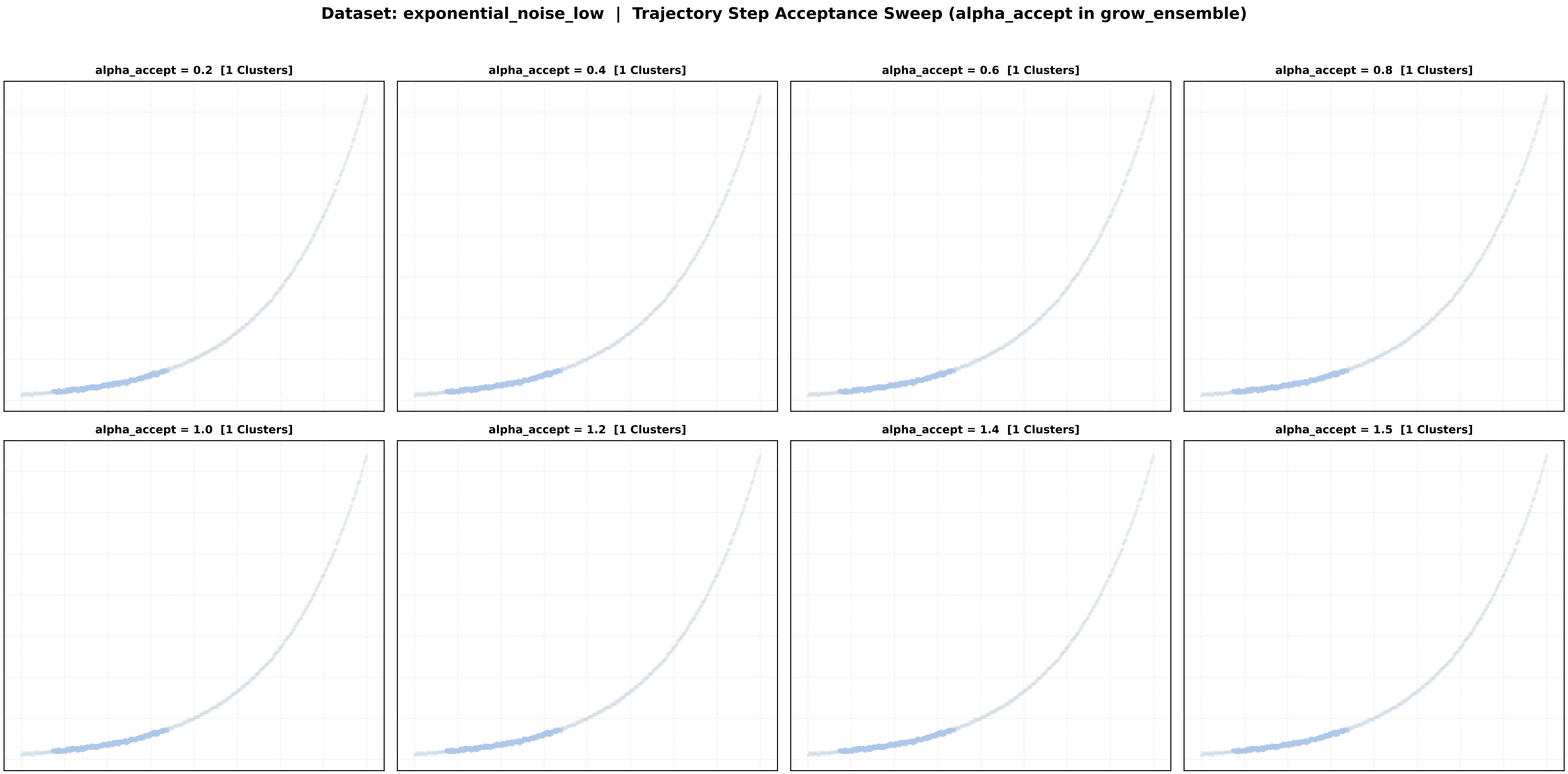



Dataset: exponential_noise_high | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [1 Clusters]
alpha_accept = 0.4 [1 Clusters]
alpha_accept = 0.6 [1 Clusters]
alpha_accept = 0.8 [1 Clusters]
alpha_accept = 1.0 [1 Clusters]
alpha_accept = 1.2 [1 Clusters]
alpha_accept = 1.4 [1 Clusters]
alpha_accept = 1.5 [1 Clusters]


Dataset: exponential_noise_low | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [1 Clusters]
alpha_accept = 0.4 [1 Clusters]
alpha_accept = 0.6 [1 Clusters]
alpha_accept = 0.8 [1 Clusters]
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alpha_accept = 1.4 [1 Clusters]
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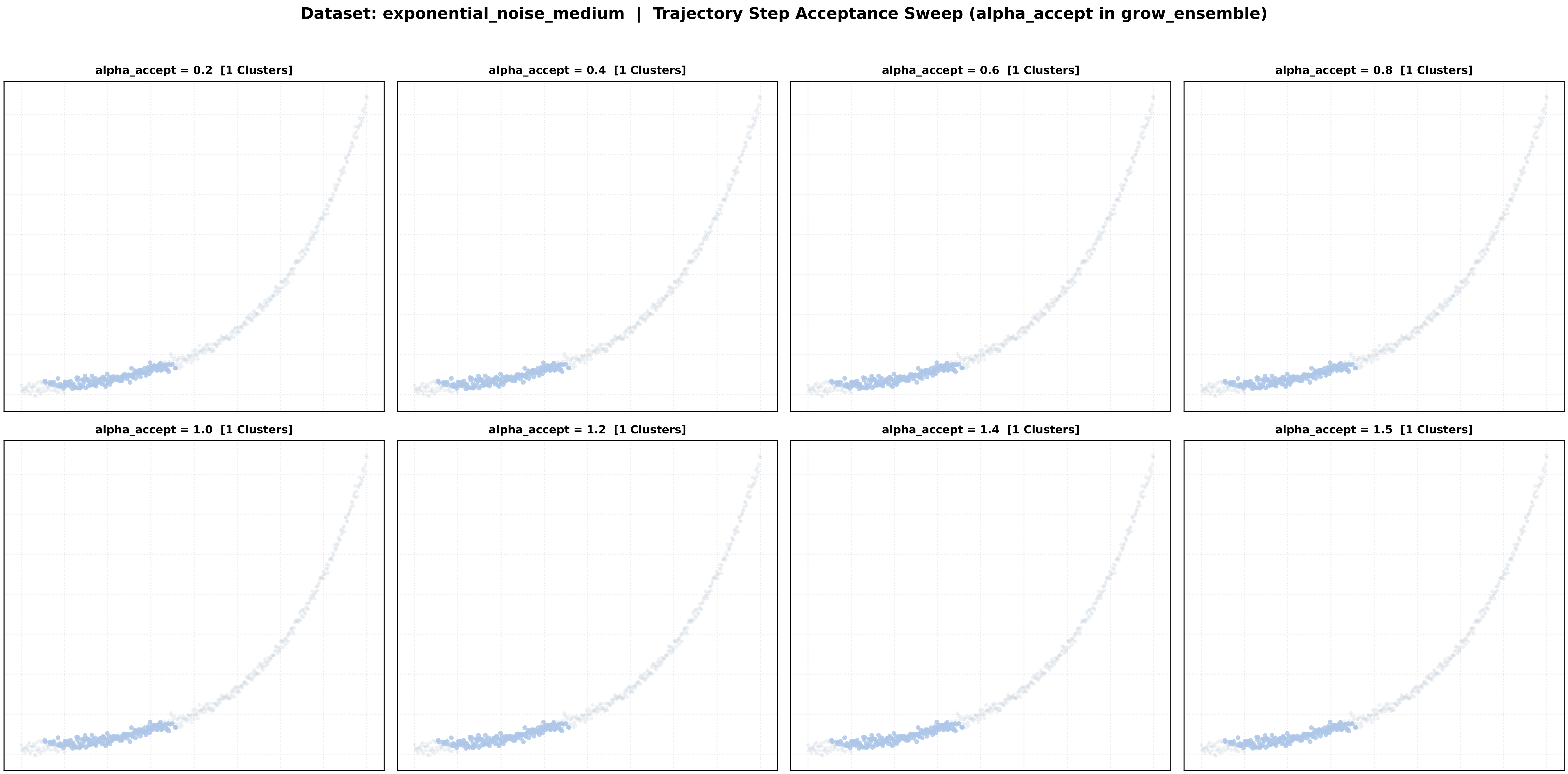
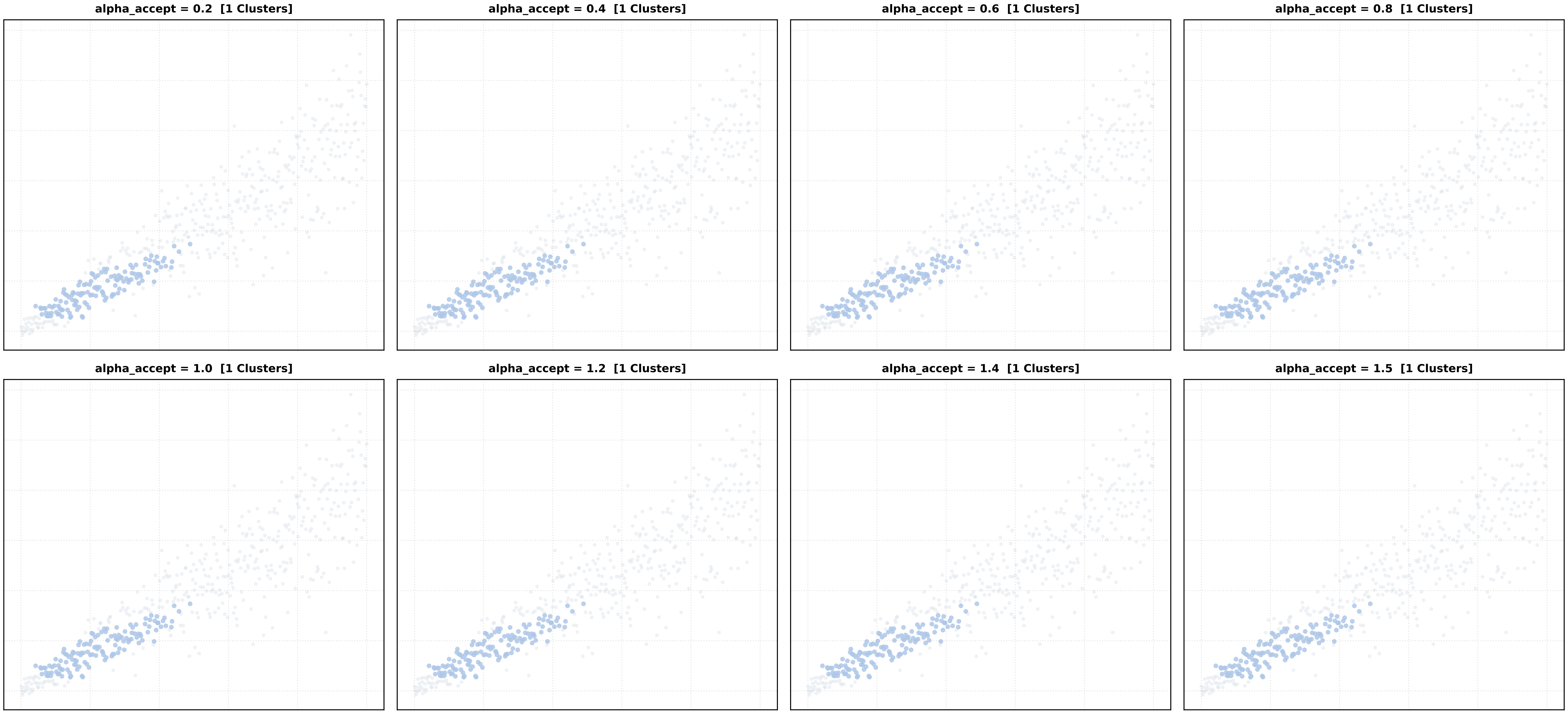
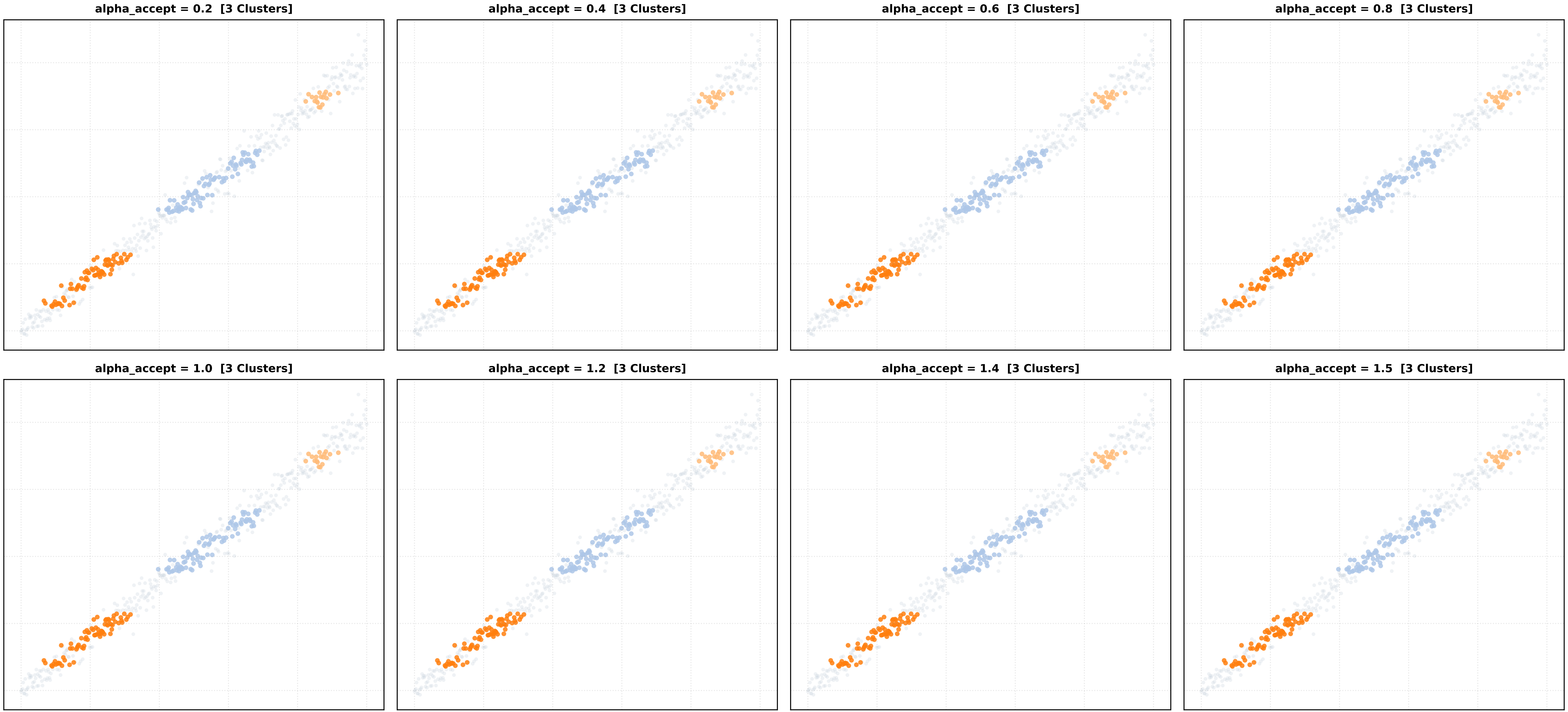

Dataset: exponential_noise_medium | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [1 Clusters]
alpha_accept = 0.4 [1 Clusters]
alpha_accept = 0.6 [1 Clusters]
alpha_accept = 0.8 [1 Clusters]
alpha_accept = 1.0 [1 Clusters]
alpha_accept = 1.2 [1 Clusters]
alpha_accept = 1.4 [1 Clusters]
alpha_accept = 1.5 [1 Clusters]

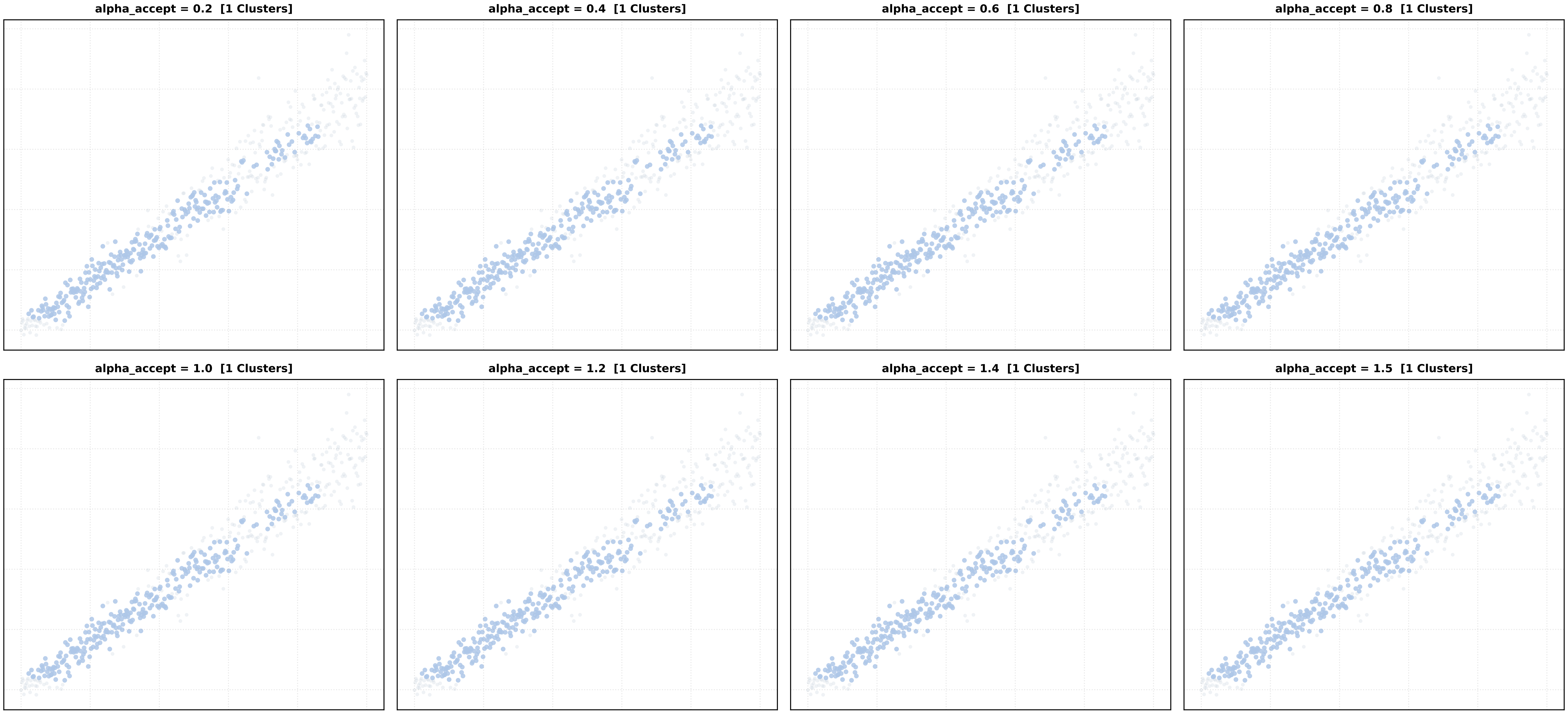
Dataset: heteroscedastic_noise_high | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)

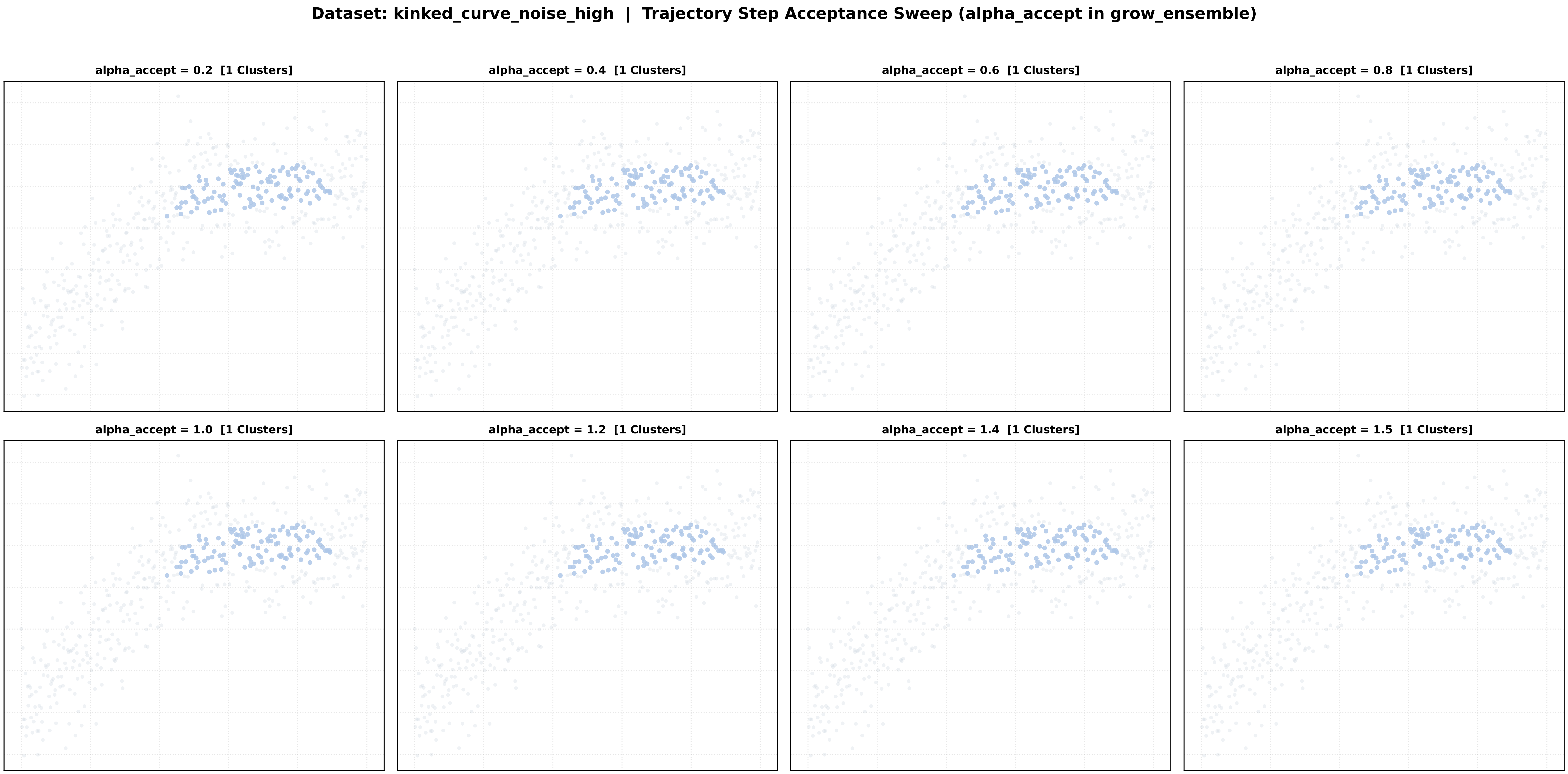


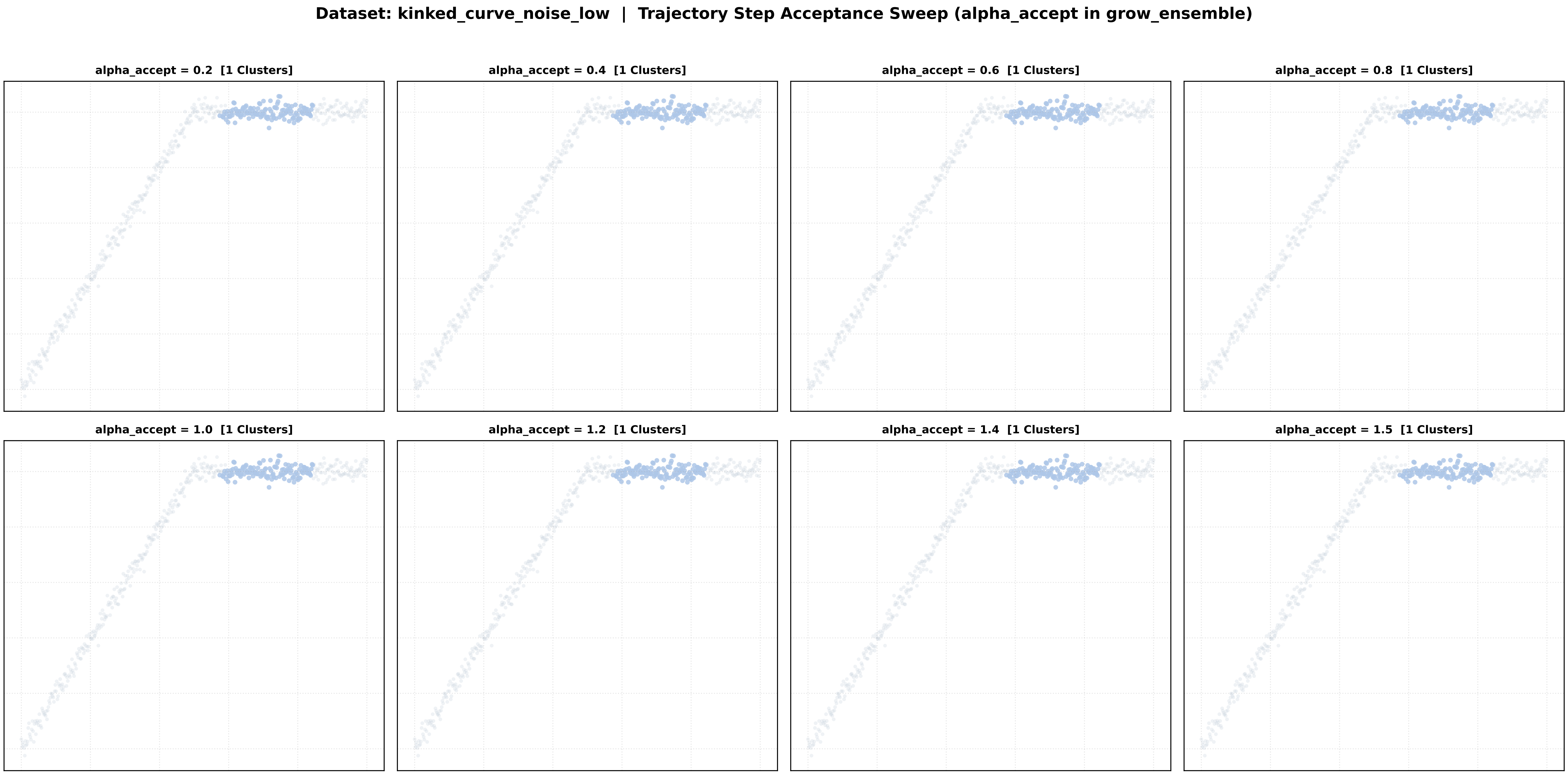
Dataset: heteroscedastic_noise_low | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)



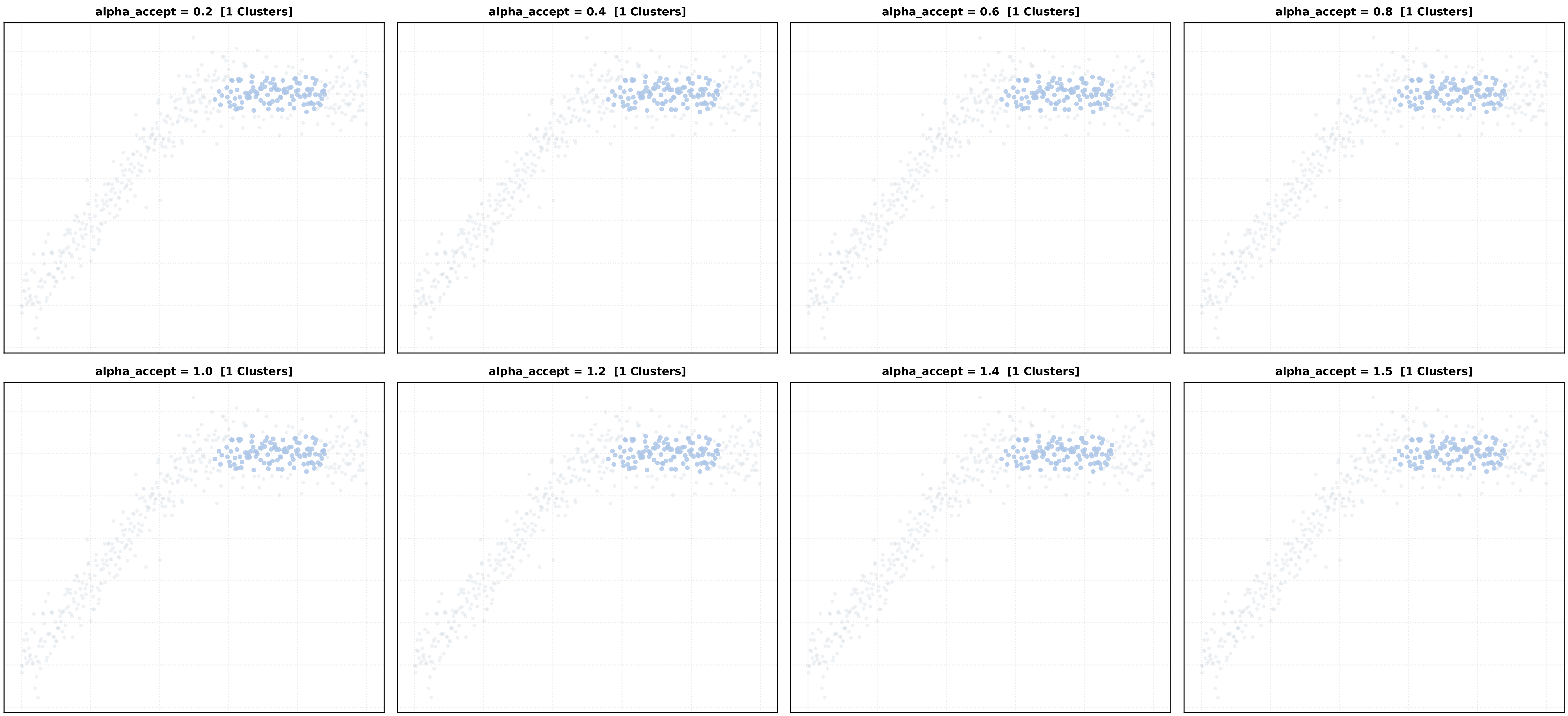
Dataset: heteroscedastic_noise_medium | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)



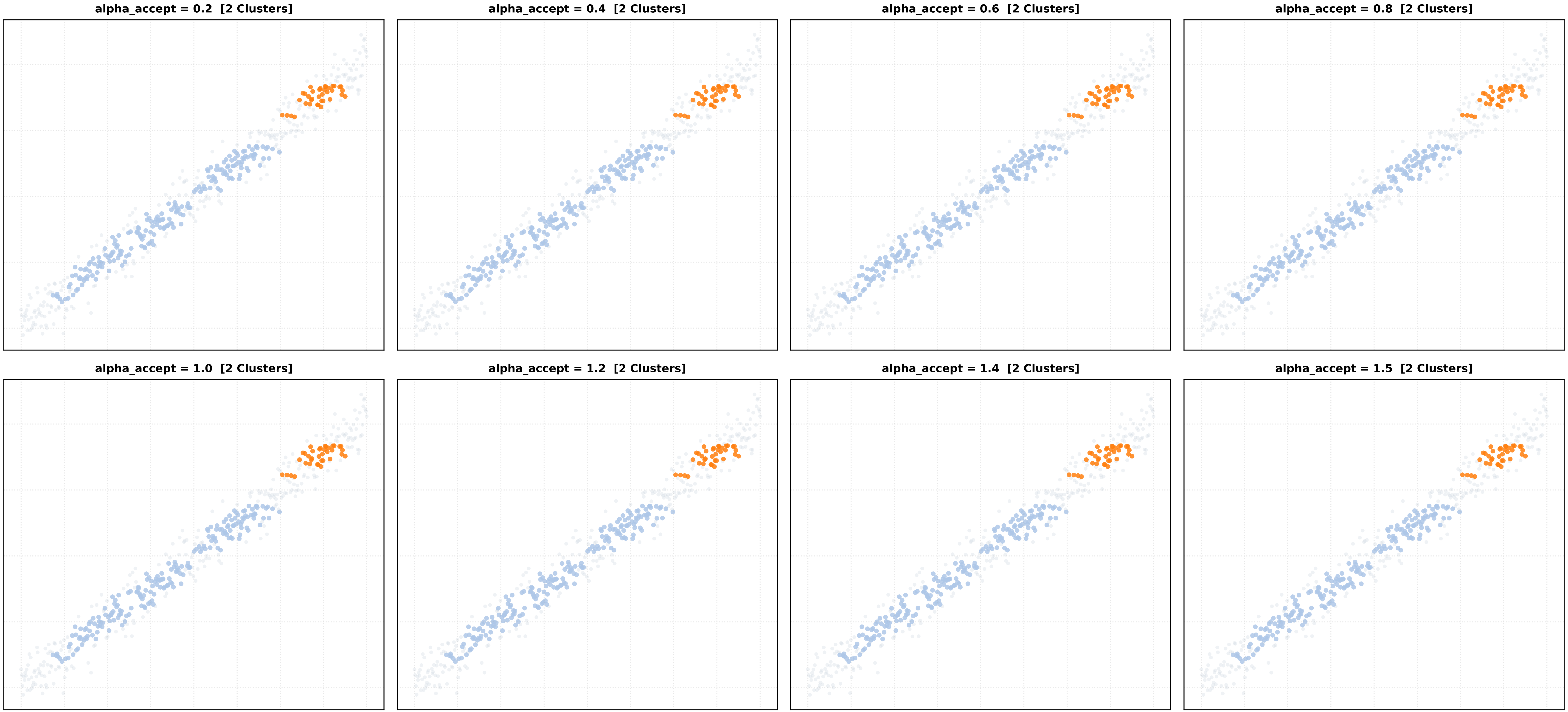

Dataset: kinked_curve_noise_high | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [1 Clusters]
alpha_accept = 0.4 [1 Clusters]
alpha_accept = 0.6 [1 Clusters]
alpha_accept = 0.8 [1 Clusters]
alpha_accept = 1.0 [1 Clusters]
alpha_accept = 1.2 [1 Clusters]
alpha_accept = 1.4 [1 Clusters]
alpha_accept = 1.5 [1 Clusters]

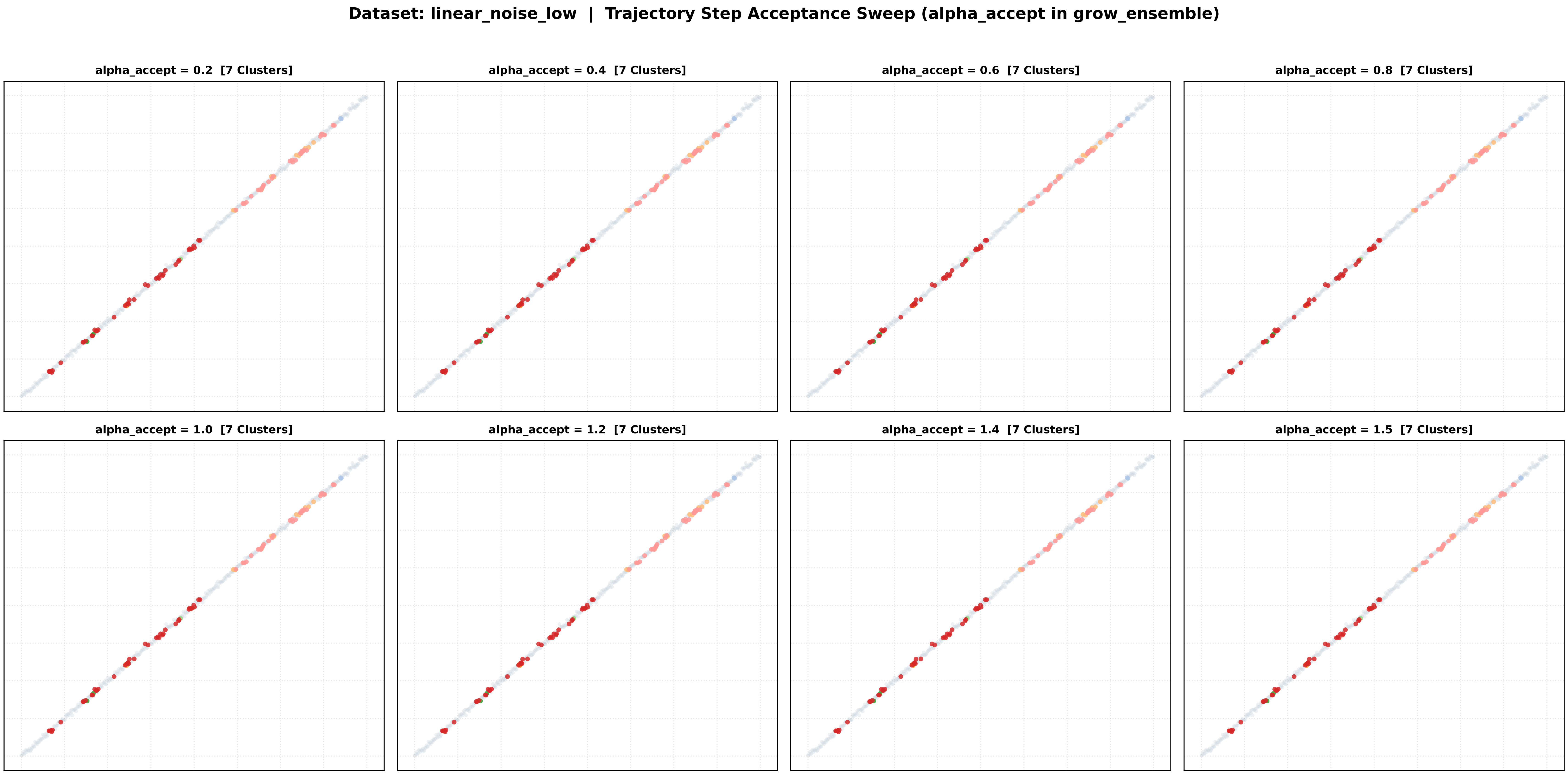

Dataset: kinked_curve_noise_low | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [1 Clusters]
alpha_accept = 0.4 [1 Clusters]
alpha_accept = 0.6 [1 Clusters]
alpha_accept = 0.8 [1 Clusters]
alpha_accept = 1.0 [1 Clusters]
alpha_accept = 1.2 [1 Clusters]
alpha_accept = 1.4 [1 Clusters]
alpha_accept = 1.5 [1 Clusters]

Dataset: kinked_curve_noise_medium | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)

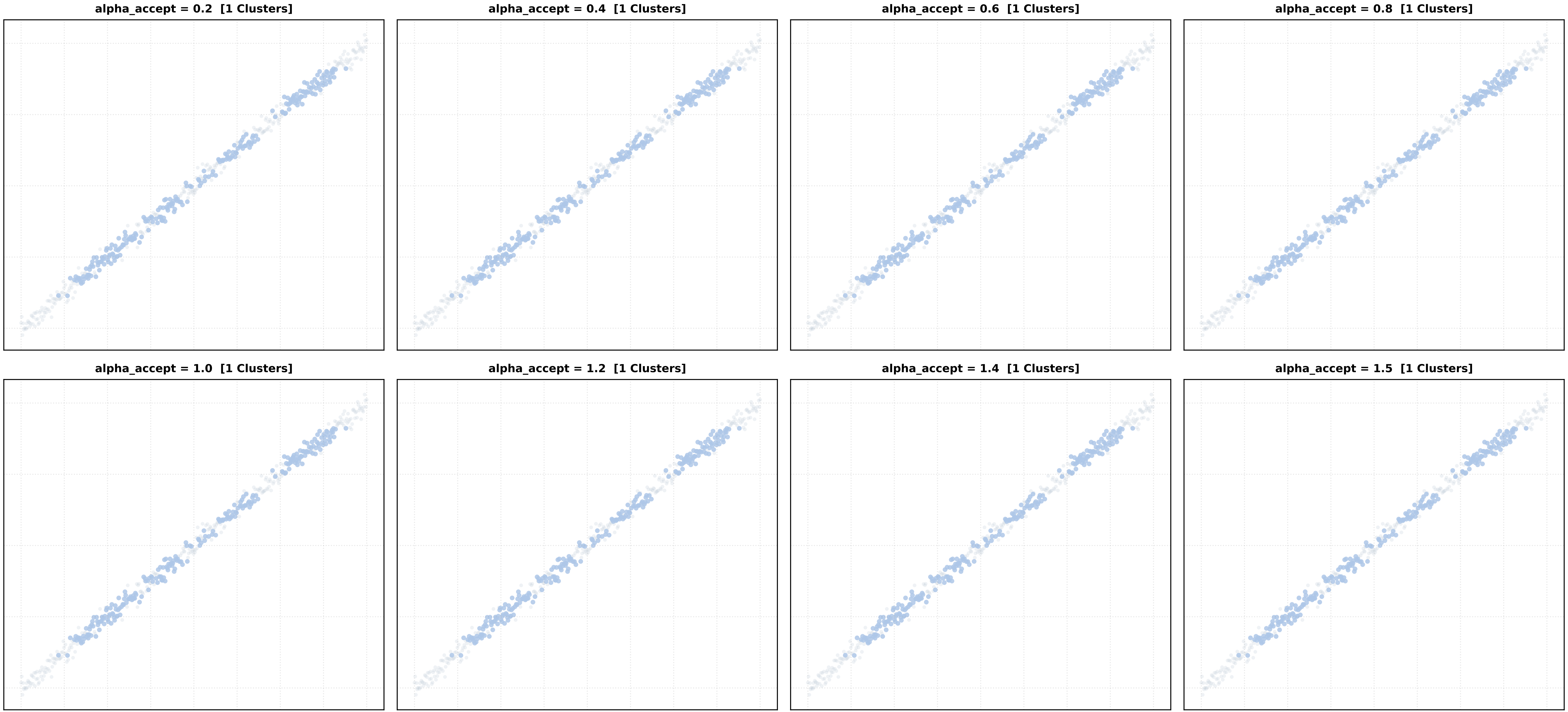


Dataset: linear_noise_high | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)

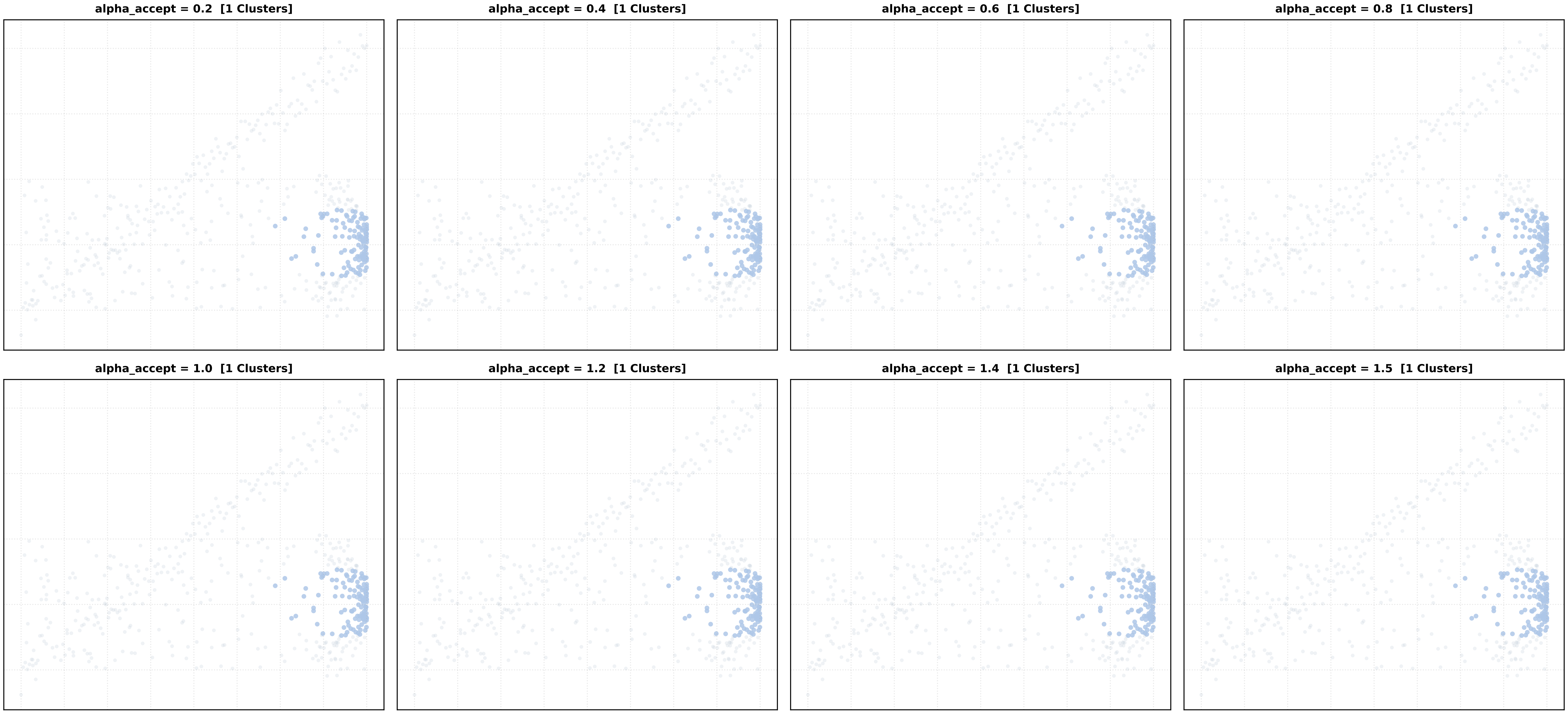


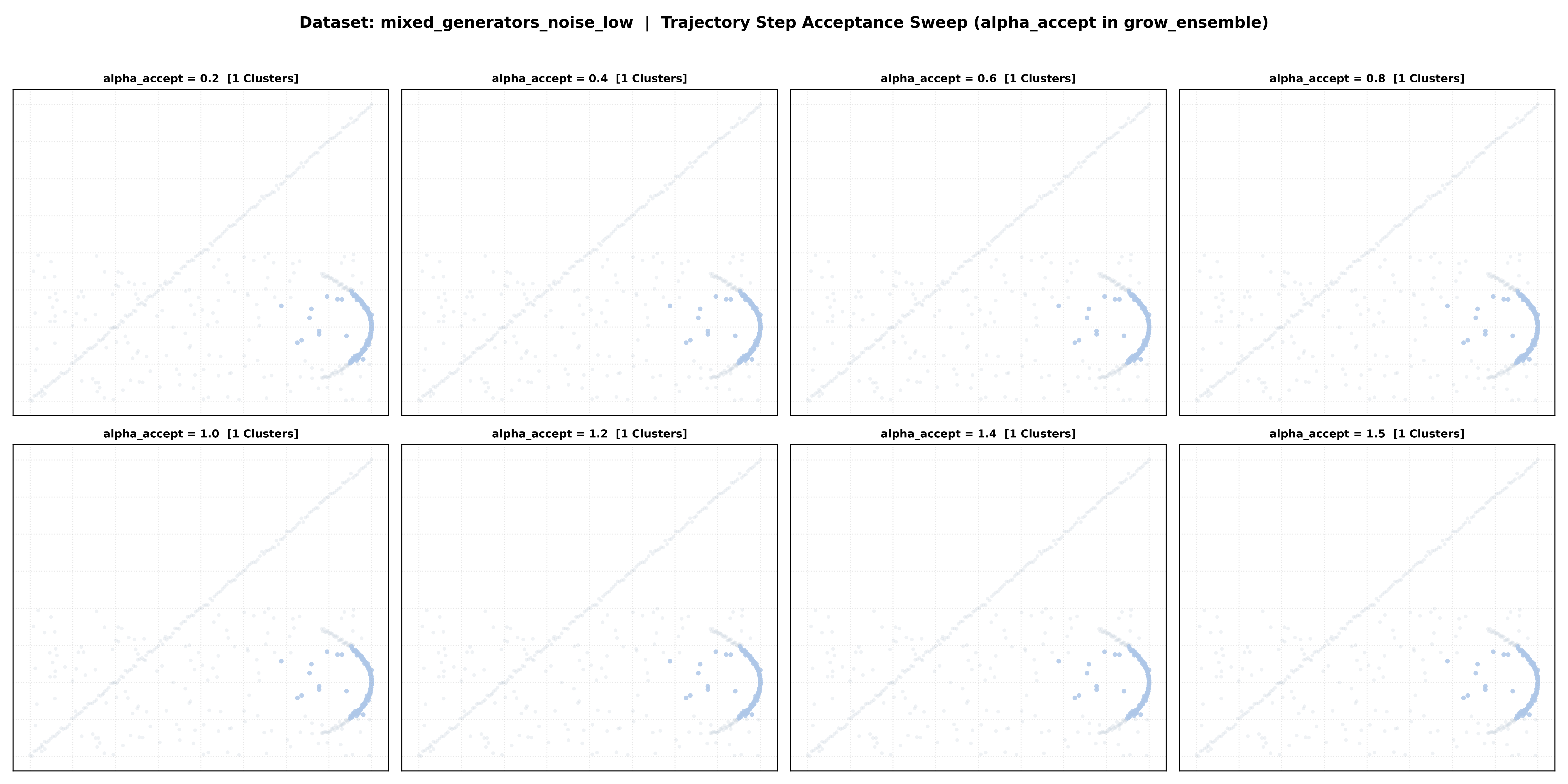

Dataset: linear_noise_low | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [7 Clusters]
alpha_accept = 0.4 [7 Clusters]
alpha_accept = 0.6 [7 Clusters]
alpha_accept = 0.8 [7 Clusters]
alpha_accept = 1.0 [7 Clusters]
alpha_accept = 1.2 [7 Clusters]
alpha_accept = 1.4 [7 Clusters]
alpha_accept = 1.5 [7 Clusters]

Dataset: linear_noise_medium | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)



Dataset: mixed_generators_noise_high | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)




Dataset: mixed_generators_noise_low | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [1 Clusters]
alpha_accept = 0.4 [1 Clusters]
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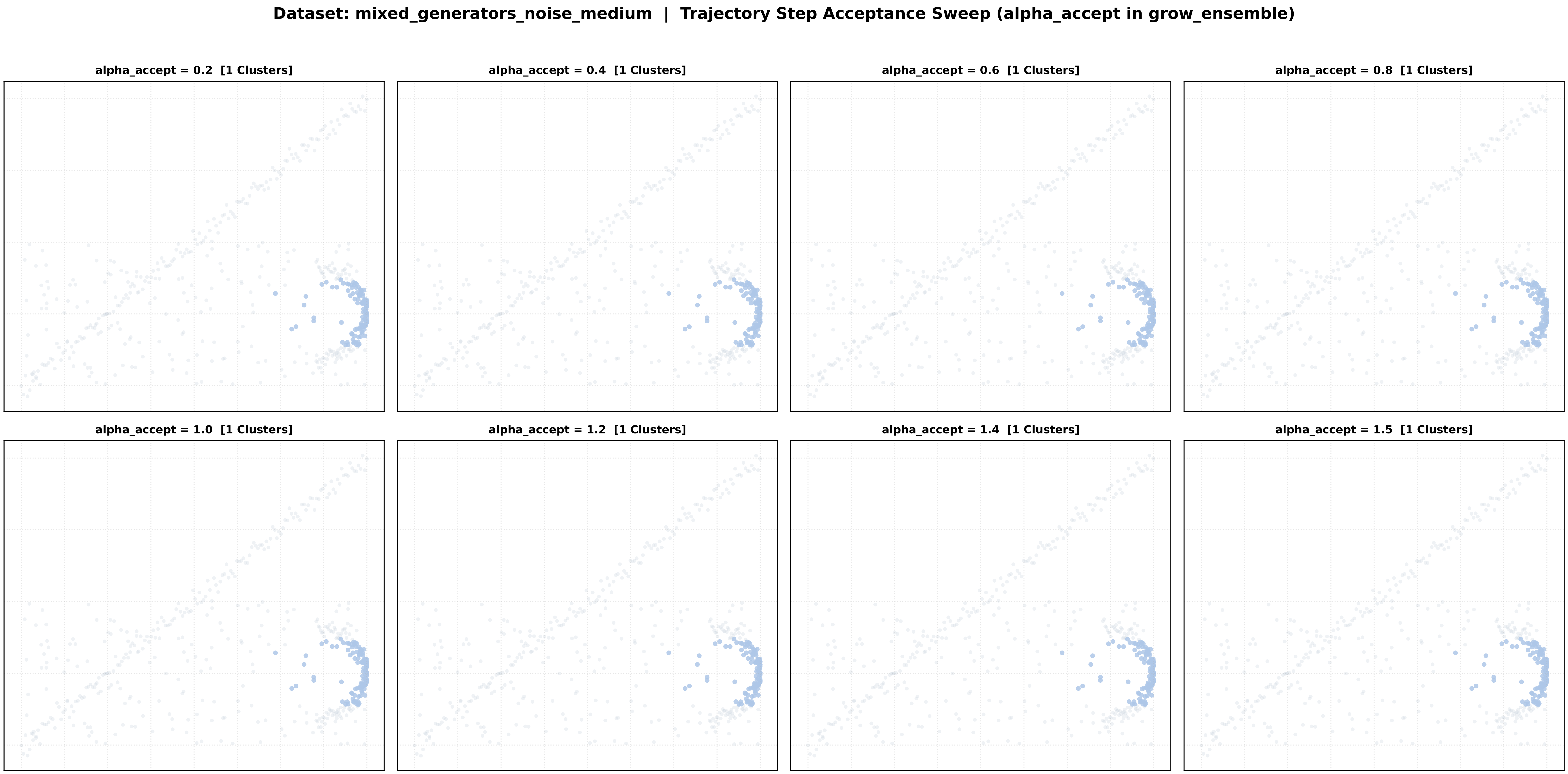
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alpha_accept = 0.2 [1 Clusters]
alpha_accept = 0.4 [1 Clusters]
alpha_accept = 0.6 [1 Clusters]
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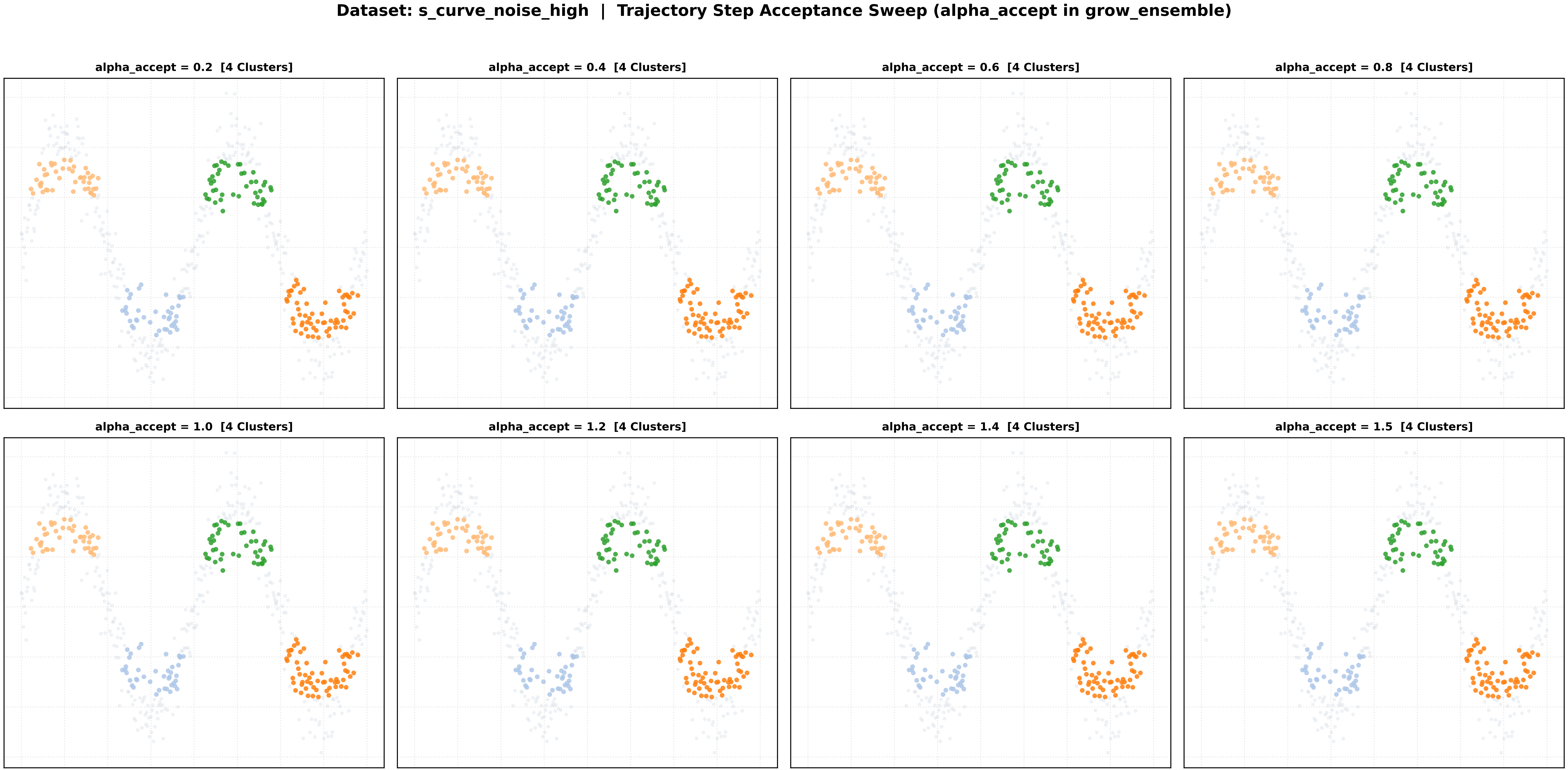
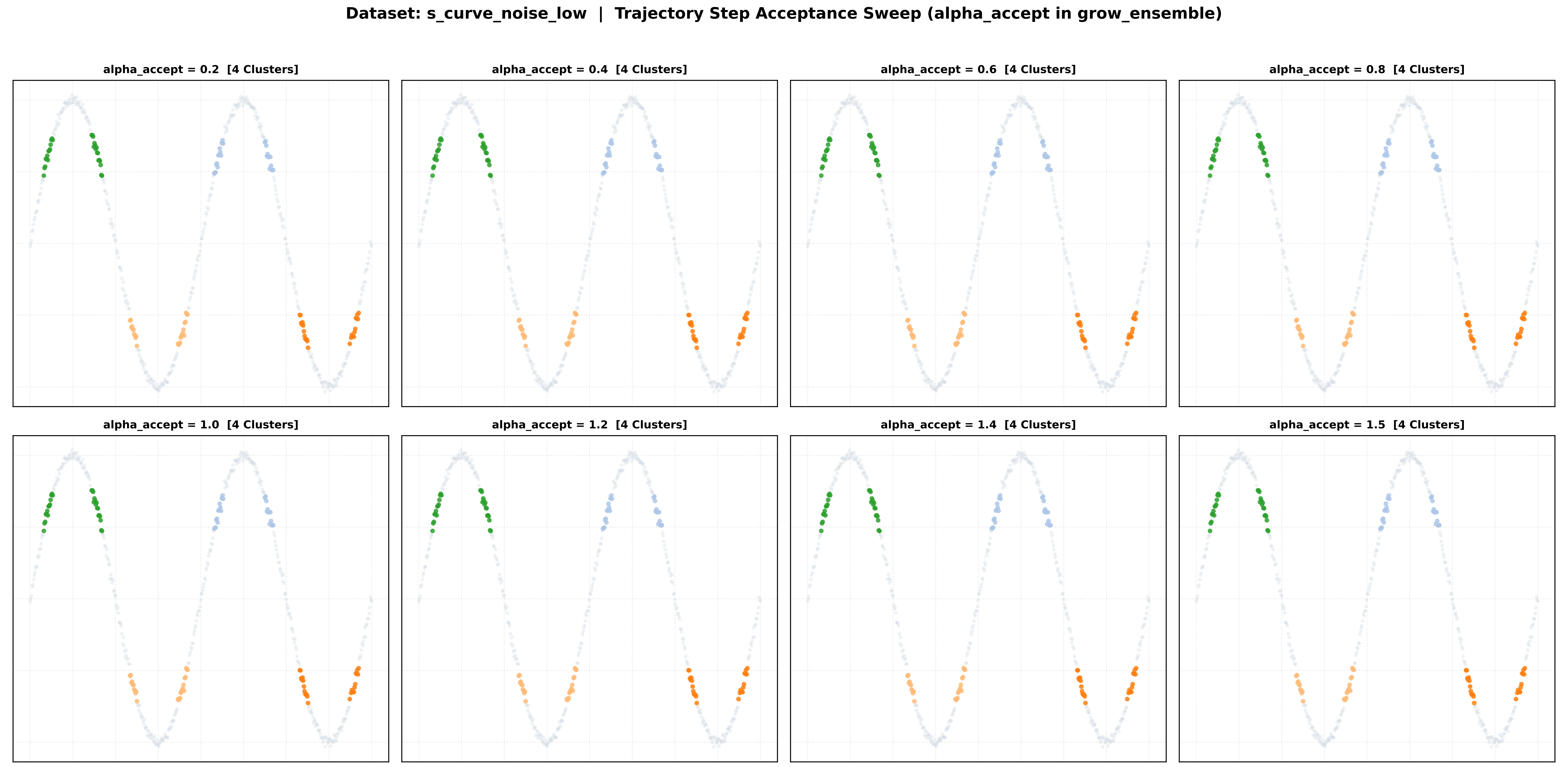

Dataset: s_curve_noise_high | Trajectory Step Acceptance Sweep (alpha_accept in grow_ensemble)
alpha_accept = 0.2 [4 Clusters]
alpha_accept = 0.4 [4 Clusters]
alpha_accept = 0.6 [4 Clusters]
alpha_accept = 0.8 [4 Clusters]
alpha_accept = 1.0 [4 Clusters]
alpha_accept = 1.2 [4 Clusters]
alpha_accept = 1.4 [4 Clusters]
alpha_accept = 1.5 [4 Clusters]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